# How Does Venture Capital Financing Improve Efficiency in Private Firms? A Look Beneath the Surface

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We use the Longitudinal Research Database (LRD) of the U.S. Census Bureau, which covers the entire universe of private and public U.S. manufacturing firms, to study several related questions regarding the efficiency gains generated by venture capital (VC) investment in private firms. First, do VCs indeed improve the efficiency (total factor productivity, TFP) of private firms, and if so, are certain kinds of VCs (high reputation vs. low reputation) better at generating such efficiency gains than others? Second, do VCs invest in more efficient firms to begin with (screening), or do they improve efficiency after investment (monitoring)? Third, do efficiency improvements due to VC backing arise from increases in sales or reductions in costs? Fourth, do VC backing and the associated efficiency gains affect the probability of a successful exit (IPO or acquisition)? Our analysis shows that the overall efficiency of VC-backed firms is higher than that of non-VC-backed firms at every point in time. This efficiency advantage of VC-backed firms arises from

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both screening and monitoring: The efficiency of VC-backed firms prior to receiving financing is higher than that of non-VC-backed firms, and further, the growth in efficiency subsequent to VC financing is greater for such firms. The above increases in efficiency of VC-backed firms are spread over the first two rounds of VC financing after which the TFP of such firms remains constant until exit. Additionally, we show that while the TFP of firms prior to receiving financing is lower for high-reputation VC-backed firms, the increase in TFP subsequent to financing is significantly greater for these firms, consistent with high-reputation VCs having greater monitoring ability. We disentangle the screening and monitoring effects of VC backing using three different methodologies: switching regression with endogenous switching, regression discontinuity analysis, and propensity score matching. We show that while overall efficiency gains generated by VC backing arise primarily from improvements in sales, the efficiency gains of high-reputation VC-backed firms arise also from lower increases in production costs. Finally, we show that VC backing and the associated efficiency gains positively affect the probability of a successful exit. (*JEL* G24)

## 1. Introduction

The role of venture capital (VC) financing in creating value for entrepreneurial firms has been widely debated in both the academic and practitioner literature. In particular, several authors in the theoretical VC literature have argued that, in addition to providing financing, VCs provide other services to private firms that can considerably enhance the probability of success of these firms (see, e.g., Berglöf 1994; Casamatta 2003; Hellmann 1998; Schmidt 2003; Ueda 2004).[1] Another theoretical literature argues that access to private financing (such as venture capital) can significantly affect the extent of innovation and therefore the productivity of a private firm when it faces competition from other firms in the product market. A prominent example of this literature is Spiegel and Tookes (2008). Their model predicts that firms will finance projects with the greatest revenue-generating ability privately (e.g., under venture financing), and will then go to the public markets only when more modest innovations remain. Practitioners also argue that in addition to providing funding for private firms, VCs contribute greatly to their success in many other ways —for example, by helping them in hiring competent management, providing better incentives to firm management and employees, as well as by allowing them access to their network of contacts among suppliers and potential customers in the product market. Further, both academics and practitioners have argued that high-reputation VCs are better at providing the above services than low-reputation VCs.[2] Despite the above large theoretical and practitioner literature that argues that VCs may create significant value in the product

1 There is also a theoretical literature that analyzes how such value creation differs across different types of VCs (e.g., Chemmanur and Chen 2003) and the relationship between the optimal size of a VC's portfolio and the VC's ability to add value to its portfolio firms (e.g., Fulghieri and Sevilir 2009).

2 See, e.g., Bygrave and Timmons (1992), who state, "It is far more important whose money you get than how much you get or how much you pay for it."

market, empirical evidence on whether VCs indeed create such value has been relatively scarce (with some exceptions that we discuss later), perhaps due to limitations on the availability of data on private firms.

This article addresses the above gap in the empirical literature by conducting a novel large sample study and analyzing the role of venture capitalists in creating productivity improvements for entrepreneurial firms using a unique database covering the universe of private firms in the U.S. manufacturing sector obtained from the Longitudinal Research Database (LRD) of the U.S. Bureau of Census. Manufacturing firms account for approximately 41% of the total number of VC-backed firms in VentureXpert and approximately 31% of all VC financing rounds between 1972 and 2000 with total VC investment in these firms being above $150 billion: Some prominent examples of manufacturing firms that have received VC financing in the past are Advanced Micro Devices (AMD), Cisco Systems, Genentech, Genzyme, Texas Instruments, Imclone, Mattel, Dell, and Sun Microsystems. Our sample of VC-backed firms is fairly representative of what we would expect of VC-backed firms (with some exceptions), with high levels of concentration in computers, biotech, electronics, and other high-tech industries such as precision instruments. The sectors that are excluded from our sample are non-packaged software, e-commerce and Internet, and other service-oriented industries such as financials. It should be noted that approximately 62% of high-tech industries, in which VCs are more inclined to invest, fall within the scope of the LRD, as these industries are part of the manufacturing sector (i.e., having Standard Industrial Classification [SIC] codes between 2000 and 3999).

We answer several interesting questions regarding the role played by VCs in creating "extra-financial" value for private firms that they invest in. We study whether VCs indeed create such value for entrepreneurial firms, and if so, how and when this value is created. We answer the following questions: First, do VC-backed firms have better performance and higher operating efficiency (as measured by total factor productivity, or TFP) than non-VC-backed firms? Second, if indeed this is the case, precisely how do VCs create value for private firms: Are they able to identify and invest in higher quality entrepreneurial firms—i.e., selection or "screening"—or does the value creation arise primarily from the various extra-financial value-added services they provide to the firm (discussed earlier) subsequent to their investing in the firm—i.e., "monitoring"? We wish to emphasize here that by monitoring we refer to both value-addition by venture capitalists, such as product market support (potentially using the VC's network of contacts) and expert advice, as well as explicit monitoring activities such as scrutinizing management actions, professionalizing management, and setting incentive compensation schemes. Third, what are the precise channels through which VCs improve the productivity of private firms? In particular, do these efficiency improvements arise from better aggregate product market performance (sales) of VC-backed firms relative to non-VC-backed firms? Or do they arise from lower production costs of VC-backed firms relative to non-VC-backed firms?[3] Fourth, if the value-addition due to VC backing is at least partly due to monitoring, how are these value improvements spread over time: Do they occur in the early stages immediately after the VCs invest in the firm, or do they occur in later stages? Fifth, are more reputable VCs capable of creating greater value by improving the efficiency of firms they invest in? In particular, are high-reputation VCs better at screening or monitoring (or both) than low-reputation VCs? Further, are the efficiency improvements arising through the various channels of sales and cost improvements greater for firms backed by high-reputation VCs than for those backed by low-reputation VCs? Finally, how do VC backing and productivity improvements affect the probability of a successful exit (IPO or acquisition) as opposed to that of a write-off?

The results of our empirical analysis can be summarized as follows. We start by investigating whether VC-backed firms are characterized by greater overall efficiency compared with non-VC-backed firms. Similar to several other papers that have used the LRD database to study various corporate events (see, e.g., Schoar 2002; Maksimovich and Philips 2008; Chemmanur, He, and Nandy 2010), we use total factor productivity (TFP) as our measure of firm efficiency. TFP growth measures the residual growth in a firm's output after accounting for the growth in output attributable to the various factors of production and the production technology in place. In other words, an increase in TFP is an increase in the overall productivity (efficiency) of the firm, since more output can now be produced than earlier, with each of the factors of production remaining the same. Venture capital financing involves an injection of additional capital into the firm, which may increase the scale of the firm and thus a mechanical increase in output and revenue. Therefore, TFP is a particularly appropriate measure to analyze the increase in firm performance due to VC backing, since it captures productivity changes after accounting for increases in the scale of production, which will not be captured by more traditional measures such as revenue or sales growth. As a robustness test, we also conduct our analysis using alternative measures of performance such as labor productivity and return on assets (ROA) and find that our results are similar to the ones we find using the TFP measure.

We find that the efficiency of VC-backed firms (as measured by TFP) prior to receiving venture financing is higher than that of non-VC-backed firms, and further, the growth in TFP subsequent to receiving venture financing is greater for VC-backed firms relative to non-VC-backed firms. We thus find evidence of both a screening and a monitoring role for VCs in improving firm efficiency. In our analysis of the dynamics of productivity growth, we first show that the above improvement in TFP of VC-backed firms relative to non-

[3] For example, such efficiency improvements could arise from a lower aggregate level of employment in venture-backed firms relative to non-venture-backed firms, or through lower salaries and wages (or both), thus leading to lower total labor costs. Alternatively, such improvements could also arise from more efficient usage of materials in the production process.

VC-backed firms increases over the four years subsequent to the year of the first round of venture financing, and remains at this higher level until exit. This is the first article to document the above finding in the literature. We also analyze the above dynamics in TFP changes round-by-round for VC-backed firms. The results of our round-by-round analysis are consistent with those discussed above and show that most of the efficiency gains for VC-backed firms are concentrated around the first couple of rounds of VC investment in the firm.

To control for the possibility that VCs may merely be providing funding to better-quality firms, which then go on to perform better subsequently (in other words, to address any potential endogeneity or selection issues related to VC backing), we employ three alternative methodologies. The first methodology we employ is “switching regressions with endogenous switching,” which controls for unobservable characteristics that affect both the probability of getting VC financing for an entrepreneurial firm as well as its productivity. The results from this regression are used to perform a counterfactual analysis that answers the following question: What would the TFP growth of VC-backed firms have been had they not received such financing (and vice versa)? Clearly, the difference between the actual TFP growth of VC-backed firms and the hypothetical level estimated from the above “what-if ” analysis yields the TFP growth attributable to the monitoring effect of VC backing. Consistent with our earlier results, our switching regression results indicate a significantly positive effect of VC monitoring on TFP growth. Specifically, we find that TFP growth would have been smaller for VC-backed firms had they not received VC financing, and TFP growth would have been greater for non-VC-backed firms had they received VC financing.

The second methodology we use to distinguish between screening and monitoring is a regression discontinuity analysis. This approach uses a discontinuous jump in the probability of obtaining VC financing to identify the causal effect of VC involvement on TFP growth. We exploit the eligibility requirements for firms to obtain financial support from the U.S. Small Business Administration (SBA) to identify a jump in the probability of VC financing, since firms that are not eligible for SBA support are more likely to seek external financing sources such as venture capital. Consistent with our previous results documenting the monitoring ability of VCs, we find a positive causal effect of VC financing on the TFP growth of private firms receiving such financing.

The third methodology we employ is a matched sample analysis using a propensity score matching algorithm.[4] Using this methodology, we match our sample of VC-backed firms to non-VC-backed private firms along the following dimensions: firm size, industry, and average TFP over the five years prior to receiving venture financing. Consistent with our earlier results,

4 While the switching regression and regression discontinuity analyses control for unobservable as well as observable differences in the quality of firms backed by VCs relative to those not backed by them, the propensity score matching methodology controls only for observable differences.

we find that the TFP growth of VC-backed firms subsequent to receiving financing is significantly greater than that of matching firms, thus confirming the monitoring effect of VC backing on TFP growth.

Our results on the channels through which VC backing improves efficiency can be summarized as follows. First, VC-backed firms are characterized by higher sales than non-VC-backed firms prior to receiving venture financing. Further, these firms are characterized by a greater increase in sales in the years subsequent to receiving venture financing compared with non-VC-backed firms. Second, total production costs are greater for VC-backed firms compared with non-VC-backed firms prior to receiving venture financing; the growth in these costs subsequent to receiving financing is also greater for VC-backed firms relative to non-VC-backed firms. Third, while total employment is similar for VC-backed and non-VC-backed firms prior to receiving venture financing, total salaries and wages are greater for VC-backed firms prior to receiving financing. Additionally, the growth in total salaries and wages subsequent to receiving financing is greater for VC-backed firms relative to non-VC-backed firms, though the growth in the level of employment remains comparable across the two kinds of firms. Overall, the above results indicate that the primary channel through which VC backing improves efficiency is by improving product market performance (sales). This increase in sales could partly be attributed to a better-quality workforce employed by VC-backed firms.

We also find that while the TFP of firms prior to VC financing is higher for low-reputation VC-backed firms, the growth in TFP subsequent to financing is higher for high-reputation VC-backed firms relative to that for low-reputation VC-backed firms. This finding suggests that while low-reputation VCs rely on selecting more efficient firms to begin with (screening), high-reputation VCs are able to improve the efficiency of the firms they invest in to a greater extent, suggesting that they have greater monitoring ability. Consistent with these results, we find significantly higher TFP at every round subsequent to receiving financing for high-reputation VC-backed firms compared with low-reputation VC-backed firms. Further, the increase in TFP in each round of VC financing (relative to the TFP before VC financing) is significantly greater for firms backed by high-reputation VCs, again suggesting greater monitoring ability for such VCs.

Our findings regarding the differences between high-reputation and low-reputation VCs in terms of the channels through which they improve the efficiency of their portfolio firms can be summarized as follows. First, the level of sales prior to receiving financing is greater for high-reputation VC-backed firms compared with low-reputation VC-backed firms, and the growth in sales subsequent to financing (in the first four years) is also greater for high-reputation VC-backed firms. Second, total production costs prior to venture financing are lower for high-reputation VC-backed firms compared with low-reputation VC-backed firms, and the growth in total production costs subsequent to financing is also lower for high-reputation VC-backed firms compared with low-reputation VC-backed firms. The above two results indicate that high-reputation VCs are able to generate greater increases in sales (output) with lower increases in production costs, thus leading to greater improvements in the TFP of firms backed by them, compared with low-reputation VCs. Overall, these results are consistent with the notion that the primary channel through which both high- and low-reputation VCs improve efficiency is through improvements in product market performance (sales); for high-reputation VCs, additional improvements in efficiency arise through reductions in input costs as well.

Finally, the results of our analysis on the impact of VC backing and efficiency improvements on the probability of a successful exit can be summarized as follows. First, VC financing has a positive association with the probability of a successful exit (either through an IPO or through an acquisition). Second, the probability of an exit through an IPO is increasing in both the average level of TFP prior to each year and TFP growth. The above results are consistent with the notion that VC backing and efficiency improvements have a long-lived impact and result in a higher likelihood of successful exit.[5] They also suggest that firms with higher levels of efficiency improvements are more likely to exit through an IPO rather than an acquisition.[6]

The rest of this article is organized as follows. Section 2 describes the data and sample selection procedures and explains the construction of various variables used in this study. Section 3 describes our empirical methodology and presents the results of our multivariate analyses relating VC backing to increases in firm efficiency. Section 4 provides a discussion of the analyses used to empirically distinguish between VC screening and monitoring using the following three methodologies: switching regression, regression discontinuity, and propensity score based matching. Section 5 analyzes the channels through which TFP and efficiency improvements are generated for VC-backed firms. Section 6 presents some of the above analyses for high- versus low-reputation VC-backed firms. Section 7 analyzes how VC backing and improvements in efficiency impact the exit decisions of firms. Section 8 discusses how our article is related to the existing literature, and Section 9 concludes.

## 2. Data, Sample Selection, and Construction of Variables

The primary data used in this study are obtained from the Longitudinal Research Database (LRD), maintained by the Center of Economic Studies at

5 In unreported tests, we also show that, within the sample of VC-backed firms, both the average TFP prior to VC backing (i.e., screening) and the TFP growth after VC backing (i.e., monitoring) positively affect the entrepreneurial firm's probability of a successful exit.

6 See Bayar and Chemmanur (2010) for a theoretical model that makes the above prediction.

the U.S. Census Bureau.[7] The LRD is a large micro database that provides plant-level information for firms in the manufacturing sector (SIC codes 2000 to 3999). In the census years (1972, 1977, 1982, 1987, 1992, 1997), the LRD covers the entire universe of manufacturing plants in the Census of Manufacturers (CM). In non-census years, the LRD tracks approximately 50,000 manufacturing plants every year in the Annual Survey of Manufacturers (ASM), which covers all plants with more than 250 employees with probability one. In addition, it also includes smaller plants that are randomly selected every fifth year to complete a rotating five-year panel. Therefore, all U.S. manufacturing plants with more than 250 employees are included in the LRD database on a yearly basis from 1972 to 2000, and smaller plants with fewer than 250 employees are included in the LRD database every census year and are also randomly included in the non-census years, continuously for five years, as a rotating five-year panel.[8] Most of the data items reported in the LRD (e.g., the number of employees, employee compensation, and total value of shipments) represent items that are also reported to the IRS, increasing the accuracy of the data.

The two major difficulties in conducting research on VC financing and its effects on firms' performance are first, obtaining firm-specific data on private firms that receive VC financing, and second, obtaining data on private firms that could potentially obtain VC financing but do not. Clearly, publicly available firm-level databases such as Compustat do not meet this criteria since it only has data on publicly traded firms. An alternate data source is another panel dataset collected by the U.S. Census Bureau, namely the Longitudinal Business Database (LBD).[9] The major advantages of using the LRD relative to the LBD for this study are the following: First, assets, sales, operating costs, and other such firm-level financial information are either not covered or mostly missing in the LBD compared with the LRD. Thus, our overall metric of firm efficiency—i.e., total factor productivity (TFP)—can be constructed only for the LRD panel. Second, the nature of the LRD allows us to identify the precise channels of value improvements in firms resulting from VC investments, which would not have been possible had we used the LBD. Due to these reasons, we focus on the LRD for the purposes of this study.

Our sample of VC investments is drawn from VentureXpert, a database maintained by Thomson Financial, which contains round-by-round information for both the firms in which VCs invest as well as the VC firms themselves.

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7 See McGuckin and Pascoe (1988), who provide a detailed description of the Longitudinal Research Database (LRD) and the method of data collection.

8 Given that a random sample of smaller plants is continuously present in our sample, our data are not substantially skewed toward larger firms; smaller firms are well represented in the data. The rotating sample of smaller plants is sampled by the Census Bureau each year in the non-census years in order to minimize such a bias in the data. We address this issue further in Panel B of Table 2, which is discussed in Section 3.

9 Similar to the LRD, the LBD is also a panel dataset that tracks the set of U.S. business establishments from 1975 to the present. While the LRD is limited to the manufacturing sector, the LBD encompasses all industry sectors. However, the LBD is not well suited for the aim of our study. We elaborate on this issue further below.

It provides information on the names and locations of VCs who invest in each round of the firm, the number of such VCs, the total amount invested per round, and also the date of each round of investment. Our initial extract from VentureXpert gives us a sample of 12,481 U.S.-based firms whose first round of VC financing lies between 1972 and 2000. This sample excludes any investment made by VC funds for buyout or acquisition purposes or where the purpose of the first round of investment was unknown or missing. We then merge this sample of firms to the Standard Statistical Establishment List (SSEL), which is a list of all business establishments in the United States maintained by the U.S. Census Bureau and updated on an annual basis.[10] We employ standard matching procedures using the names and addresses of firms that are commonly used by U.S. Census Bureau researchers and those working with these databases, which yields a positive match for 10,355 firms, giving us a match rate of about 83%.[11] We then restrict the sample to the manufacturing sector (SIC codes 2000 to 3999), and merge to the LRD. We keep only those firms for which we have detailed information to calculate TFP at the four-digit SIC and annual level, which leaves us with a final sample of 1,881 VC-backed firms representing 16,824 firm-years of data. This sample represents approximately 36% of VC-backed manufacturing firms from VentureXpert that are matched to the SSEL.[12] Alternatively, assuming that the proportion of manufacturing firms in VentureXpert is similar to that in the SSEL, our sample of 1,881 U.S. manufacturing firms represents approximately 30% of all U.S. manufacturing firms that received VC financing between 1972 and 2000. It is also worth pointing out that our sample contains roughly 15% of all U.S. VC-backed firms (manufacturing as well as non-manufacturing) that received VC financing between 1972 and 2000. Overall, while our sample is



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10 The SSEL is the Business Register or the "master" dataset of the U.S. Census Bureau from which both the LRD and the LBD are constructed. The SSEL contains data from the U.S. government administrative records, such as tax returns, and is augmented with data from various Census surveys. The SSEL data are at the establishment level—an establishment is a single physical location where business is conducted. The SSEL provides names and addresses of establishments and also numerical identifiers at both the establishment level as well as the firm level, through which one can link the SSEL to the LRD. Both the SSEL and the LRD provide a permanent plant number (PPN) and a firm identifier (FID), both of which remain invariant through time. We use these identifiers to track the plants and the firms forward and backward in time. A good description of the SSEL can be found in Jarmin and Miranda (2002).

11 This match rate is comparable to that achieved by earlier studies, such as Chemmanur, He, and Nandy (2010) and Puri and Zarutskie (2009), who provide a detailed description of the matching procedures employing name and address matching.

12 In unreported tests, we compare our LRD matched sample to the rest of the VC-backed manufacturing sample, based on information available from *VentureXpert*. We find that our sample is fairly representative of manufacturing firms in *VentureXpert*. In particular, our sample of LRD matched firms receives similar amounts of VC funding (first round as well as total funding) over a similar number of rounds compared with non-LRD matched firms. One difference is with respect to the proportion of early stage investments: The LRD matched sample has about 42% of firms receiving early-stage investments, while the non-LRD matched sample of manufacturing firms has 60% of firms receiving early-stage investments. We therefore provide a robustness check of our results by separately conducting our analysis for early- and late-stage firms. Please see Table 4 for this analysis.

somewhat tilted toward large, later-stage firms, it is fairly representative of U.S. manufacturing firms that received VC financing during the 1972–2000 period.

Panel A of Table 1 presents the industry distribution at the two-digit SIC level of the firms that received VC financing in our sample, while Panel B presents the number of firms that received their first round of VC financing in any given year over our sample period. As can be seen from this table, our matched sample of VC-backed firms is fairly representative of what we would expect of venture capital firms (with some exceptions), with high levels of concentration in computers, biotech, electronics, and other high-tech industries, such as precision instruments. However, non-packaged software, e-commerce and Internet, and other service-oriented firms, such as financial firms, are not included in our sample. It should be noted that approximately 62% of the high-tech industries, including computers, telecom, biotech, and

**Table 1**
**Industry and year distribution of VC-backed firms matched to LRD**

| *Panel A: Industry Distribution* | | | | *Panel B: Year Distribution* | | |
|---|---|---|---|---|---|---|
| 2-Digit SIC Code | Industry Description | Freq. | Percentage | Year of First VC Financing | Freq. | Percentage |
| 20 | Food and kindred products | 61 | 3.24 | 1972 | 18 | 0.96 |
| 22 | Textile mill products | 30 | 1.59 | 1973 | 29 | 1.54 |
| 23 | Apparel and other textile products | 35 | 1.86 | 1974 | 16 | 0.85 |
| 24 | Lumber and wood products | 25 | 1.33 | 1975 | 24 | 1.28 |
| 25 | Furniture and fixtures | 18 | 0.96 | 1976 | 24 | 1.28 |
| 26 | Paper and allied products | 32 | 1.7 | 1977 | 30 | 1.59 |
| 27 | Printing and publishing | 96 | 5.1 | 1978 | 53 | 2.82 |
| 28 | Chemicals and allied products (Biotech) | 95 | 5.05 | 1979 | 46 | 2.45 |
| 29 | Petroleum and coal products | 10 | 0.53 | 1980 | 71 | 3.77 |
| 30 | Rubber and miscellaneous plastics products | 68 | 3.62 | 1981 | 102 | 5.42 |
| 31 | Leather and leather products | 25 | 1.33 | 1982 | 101 | 5.37 |
| 32 | Stone, clay, and glass products | 67 | 3.56 | 1983 | 119 | 6.33 |
| 33 | Primary metal industries | 227 | 12.07 | 1984 | 94 | 5 |
| 34 | Fabricated metal products | 82 | 4.36 | 1985 | 78 | 4.15 |
| 35 | Industrial machinery and equipment (Computers) | 320 | 17.01 | 1986 | 85 | 4.52 |
| 36 | Electronic and other electric equipment (Telecom) | 355 | 18.87 | 1987 | 69 | 3.67 |
| 37 | Transportation equipment | 45 | 2.39 | 1988 | 83 | 4.41 |
| 38 | Instruments and related products | 247 | 13.13 | 1989 | 86 | 4.57 |
| 39 | Miscellaneous manufacturing industries | 43 | 2.29 | 1990 | 55 | 2.92 |
| | | | | 1991 | 24 | 1.28 |
| | | | | 1992 | 42 | 2.23 |
| | | | | 1993 | 34 | 1.81 |
| | | | | 1994 | 31 | 1.65 |
| | | | | 1995 | 76 | 4.04 |
| | | | | 1996 | 69 | 3.67 |
| | | | | 1997 | 93 | 4.94 |
| | | | | 1998 | 145 | 7.71 |
| | | | | 1999 | 94 | 5 |
| | | | | 2000 | 90 | 4.78 |

This table reports the distribution of two-digit SIC industry codes and the year of the first round of VC financing for U.S. VC-backed manufacturing firms from the VentureXpert database matched to the LRD. Any investment made by VC funds for buyout or acquisition purposes or where the purpose of the first round of investment was unknown or missing in the VentureXpert database was removed from the sample prior to the matching. The sample period is from 1972 to 2000.

other industries, in which VCs are more inclined to invest (as anecdotal evidence suggests), fall within the scope of the LRD, as these industries are part of the manufacturing sector—i.e., having 4-digit SIC codes between 2000 and 3999. In addition, consistent with anecdotal evidence, one can also observe that VC investment in new firms increased during the Internet bubble period of the late 1990s.

Furthermore, since the objective of our article is to analyze the impact of VC investments in private entrepreneurial firms, we also identify all public firms (as defined by CRSP) for every year in our sample and remove them from the LRD by using a similar matching approach as before. Thus, in any given year within our sample, we are left with only private firms, all of which could potentially receive VC funding, giving us a sample of 185,882 non-VC-backed firms, representing 771,830 firm-years of data.[13,14]

### 2.1 Measurement of total factor productivity (TFP)

The primary measure of firm efficiency used in our analysis is total factor productivity (TFP), which is calculated from the LRD for each individual plant at the annual four-digit (SIC) industry level as in Chemmanur, He, and Nandy (2010). Firm-level TFP is then calculated as a weighted sum of plant TFP for each year. Increasingly, several articles in the finance and economics literature have used TFP to measure firm efficiency; see, e.g., Schoar (2002), Maksimovich and Philips (2008), Maksimovich and Philips (2001), McGuckin and Nguyen (1995), Lichtenberg and Siegel (1990), and Harris, Siegel, and Wright (2005), among others. We obtain measures of TFP at the plant level, by estimating a log-linear Cobb-Douglas production function for each industry and year. Industry is defined at the level of four-digit SIC codes.[15] Individual plants are indexed $i$, industries $j$, for each year $t$ in the sample:

$$\ln\left(Y_{ijt}\right) = \alpha_{jt} + \beta_{jt}\ln\left(K_{ijt}\right) + \gamma_{jt}\ln\left(L_{ijt}\right) + \delta_{jt}\ln\left(M_{ijt}\right) + \varepsilon_{ijt}. \qquad (1)$$

We use the LRD data to construct as closely as possible the variables in the production function. Output (Y) is constructed as plant sales (total value of

[13] Note that some public firms may reenter our sample if they went through a leveraged buyout (LBO) or a management buyout (MBO) or otherwise became private again. As mentioned above, we remove any firms that received VC funding for the primary reason of acquisition or buyout. Thus, if any of these firms received VC funding during the process of becoming private, then they are eliminated from our data; if, in contrast, they were not involved in a buyout with funding from VCs, we retain them in the data.

[14] It should be noted that both the SSEL and the LRD provide establishment-level (i.e., plant-level) data. For the purpose of our analysis, we aggregate these data to the firm level using standard techiniques used in the literature previously (for example, see Chemmanur, He, and Nandy 2010) and numerical identifiers for plants and firms provided in the LRD, which we discuss further below.

[15] As a robustness check, we reestimate the production function in several different ways. First, we use two- and three-digit SIC industry classifications. Second, we estimate TFP with value-added production function specifications and separate white- and blue-collar labor inputs. Third, we divide each annual four-digit SIC industry into two groups based on capital intensity—i.e., plants with capital intensity greater than the median capital intensity for that annual four-digit group are put in one group, while those with capital intensity less than the median are put in another group. We then estimate the production function for each group separately. In all cases, we find qualitatively equivalent results.

shipments in the LRD) plus changes in the value of inventories for finished goods and work-in-progress.[16] Since we appropriately deflate plant sales by the annual industry-specific price deflator, our measure should be proportional to the actual quantity of output.[17]

Labor input (L) is defined as production-worker-equivalent man-hours—that is, the product of production-worker man-hours, and the ratio of total wages and salaries to production-worker wages. We also reestimate the TFP regression by specifying labor input to separately include non-production workers, which yields qualitatively similar results. Values for capital stock (K) are generated by the recursive perpetual inventory formula. We use the earliest available book value of capital as the initial value of net stock of plant capital (this is either the value in 1972, or the first year a plant appears in the LRD sample). These values are written forward annually with nominal capital expenditure (appropriately deflated at the industry level) and depreciated by the economic depreciation rate at the industry level obtained from the Bureau of Economic Analysis. Since values of all these variables are available separately for buildings and machinery, we perform this procedure separately for each category of assets. The resulting series are then added together to yield our capital stock measure.

Finally, material input (M) is defined as expenses for the cost of materials and parts purchased, resales, contract work, and fuel and energy purchased, adjusted for the change in the value of material inventories. All the variables are deflated using annual price deflators for output, materials, and investment at the four-digit SIC level from the Bartelsman and Gray National Bureau of Economic Research (NBER) Productivity Database.[18] Deflators for capital stock are available from the Bureau of Economic Analysis.[19] Plant-level TFP is then computed as the residuals of regression (1), estimated separately for each year and each four-digit SIC industry. Therefore, the average TFP (i.e., the average of the residuals) in any four-digit SIC industry-year is zero by construction. Plant-level TFP measures are then aggregated to the firm level by a value-weighted approach, where the weight on a plant is the ratio of its output (total value of shipments) to the total output of the firm.[20] The firm-level TFP is then winsorized at the 1st and 99th percentiles.

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16 More accurately, we use log of one plus revenue and cost measures so as not to exclude firms that have zero values for these variables.

17 Thus, the dispersion of TFP for firms in our sample should almost entirely reflect dispersions in efficiency. For the purposes of this study, however, it does not matter even if a portion of the change in TFP arises from changes in prices for VC-backed firms. For example, VCs may be able to obtain higher prices for the products of the firms they invest in, compared with those of non-VC-backed firms. However, even if an increase in price partially leads to an increase in TFP for VC-backed firms, such an increase in TFP can still be considered an extra-financial value-added service (monitoring) that VC-backed firms obtain due to their affiliation with the VC.

18 See Bartelsman and Gray (1996) for details.

19 See Lichtenberg (1992) for a detailed description of the construction of TFP measures from LRD variables.

20 As a robustness check, we also used the ratio of its capital stock to the total capital stock of the firm and the ratio of plant employment to firm employment as weights. In all cases, our results remain qualitatively unchanged.

### 2.2 Intuitive interpretation, advantages, and limitations of TFP

TFP growth measures the residual growth in a firm's output after accounting for the growth in output attributable to the various factors of production and the production technology in place. In other words, an increase in TFP is an increase in the overall productivity (efficiency) of the firm, since more output can now be produced than earlier, with each of the factors of production remaining the same. Intuitively, TFP can be understood as the difference between the actual output produced by a plant compared with its "predicted output." This "predicted output" is what the plant would have produced, given the amount of inputs it used and the industry production technology in place. Hence, a plant that produces *more* than the predicted amount of output in any given year has a greater than average productivity for that year, whereas a plant that produces *less* than the predicted amount of output in any given year has a lower than average productivity for that year. Thus, TFP can be interpreted as the relative productivity rank of a plant within its industry in any given year. A *positive* TFP implies that the plant is *above average* within its industry in that year, while a *negative* TFP implies that the plant is *below average* within its industry in that year. Since these regressions include a constant term, TFP contains only the idiosyncratic part of plant productivity. Note that firms having zero output will also be in our sample; these firms will have negative TFP.

By now, TFP has been used in the finance and economics literature as the appropriate measure of firm efficiency in various contexts. In particular, there is an established literature on corporate restructuring that relies on TFP measures (see, e.g., Schoar 2002; Maksimovich and Philips 2008; McGuckin and Nguyen 1995; Lichtenberg and Siegel 1990; Maksimovich and Philips 2001; Harris, Siegel, and Wright 2005). Similar to entrepreneurial firms, many firms undergoing restructuring (especially leveraged buyouts) also go through a number of years in which costs exceed revenues (i.e., these firms have zero or negative profits). While the cost structure in restructuring is not identical to the case of entrepreneurial firms, the underlying logic is similar. In terms of using TFP as an efficiency measure to study young entrepreneurial firms, the article that is closest to ours is Chemmanur, He, and Nandy (2010), who use TFP as their primary measure of firm efficiency in analyzing the going-public decision of private firms and how it is related to their product market characteristics.

TFP has many advantages over other measures in capturing the efficiency of firms. In particular, venture capital financing involves the injection of additional capital into the firm, which may increase the scale of the firm and thus a mechanical increase in output. Therefore, TFP is a particularly appropriate measure to analyze the increase in firm performance due to VC backing, since it captures productivity changes after accounting for increases in the scale of production, which will not be captured by more traditional measures such as revenue or sales growth. TFP is also more flexible than cash-flow measures of performance, since it does not impose the restriction of constant returns to scale and constant elasticity of scale. Also, since coefficients on capital, labor, and material inputs can vary by industry and year, this specification allows for different factor intensities in different industries in different years. One significant benefit of this is that it allows for changes in technology in each industry over time.

Of course, the TFP measure also has some limitations. In particular, TFP is not an accounting measure, and we cannot directly link the TFP of a given firm to its profitability without additional assumptions. However, Schoar (2002) suggests a way of linking TFP to profits, which is as follows. Holding input costs constant, a certain percentage increase in productivity translates to an equal percentage increase in revenues, *ceteris paribus*. An increase in revenues leads to a more than proportional increase in profits, since the elasticity of profits to productivity is greater than one. Intuitively, an increase in productivity holding all else constant leads to higher revenues without changing costs. Since profits are revenues minus costs, the smaller the profit margin, the higher the elasticity of profits to productivity.

### 2.3 Other variables used in our analysis

In this subsection, we discuss the construction and measurement of the different firm-specific variables as well as other proxies used in our analysis. The LRD contains detailed information at the plant level on the various production function parameters, such as total value of shipment, employment, labor costs, material costs, new capital investment for the purchase of buildings, machinery, equipment, etc. Using this detailed information, we first construct the variables of interest at the plant level, and then aggregate the plant level information to firm-level measures.

*Capital Stock* is constructed via the perpetual inventory method, discussed in Section 2.1. We measure *Firm Age* as the natural logarithm of one plus the number of years since the first plant of the firm first appeared in the LRD.[21] *Sales* is defined as the total value of shipment in thousands of dollars. *Capital Expenditure* is the dollar value the firm spends on the purchase and maintenance of plant, machinery, and equipment, etc. *Material Cost* is the expense for the cost of materials and parts purchased, resales, contract work, and fuel and energy purchased. *Rental and Administrative Expenditure* is the rental payments or equivalent charges made during the year for the use of

[21] In order to properly construct the age variable for plants, we start from the Census of 1962, which is the first year for which data are available from the Census Bureau. For plants that started prior to 1962, we use 1962 as the first year for that plant. The age of the firm is then set equal to the age of the first plant of the firm. Given the sampling scheme and scope of LRD, this measure is highly correlated with the actual age of the firm, which we can verify for our sample of VC-backed firms, since we are able to obtain the founding date of the VC-backed firms from the Securities Data Company (SDC) database. Particularly, the relative age between venture- and non-venture-backed firms, which is more relevant for our analysis, is captured very well by this measure.

buildings, structures, and various office equipment. *Salaries and Wages* is the total production-worker wages plus total non-production-worker wages plus total supplemental labor costs, which include both legally required supplemental labor costs as well as voluntary supplemental labor costs of the firms. *Total Production Cost* is calculated as the sum of materials cost plus rental and administrative expenditures plus salaries and wages. All the dollar values in the LRD are in thousands of dollars (in 1998 real terms), and all the plant-level measures are winsorized at the 1st and 99th percentiles.

We define *Firm Size* as the natural logarithm of capital stock of the firm. *Market Share* is defined as the firm's market share in terms of sales at the annual three-digit SIC level. We use the market share of the firm to proxy for the firm's industry leader position. We construct the industry *Herfindahl Index* based on the market share measure of each firm in the LRD. The Herfindahl index is calculated by summing up the square of each firm's market share (in sales) at the annual three-digit SIC level. A higher Herfindahl index means that the industry is more concentrated. *Industry Risk* is defined as the standard deviation of total firm sales calculated over the prior five-year period in the same three-digit SIC industry. We define *High Tech Firms* as firms belonging to the following three-digit SIC codes: 357, 366, 367, 372, 381, 382, and 384. We also control for the *Number of Plants* in a firm defined as the number of plants belonging to the firm in that particular year. In order to control for overall equity market conditions, we use *S&P 500 Returns,* which is defined as the annual return on the Standard & Poor's 500 Index.

In each year, we define VC reputation using the market share of the amount of funds raised by the VC over the prior five-year rolling window, following Megginson and Weiss (1991). We think this measure accurately captures VC reputation, since reputation is primarily built on past success and VCs will be able to raise greater follow-on funds only if the performance of their prior existing funds has been successful. This is because limited partners will be more willing to invest money when a fund has been more successful in the past. In a recent article, Hochberg, Ljungqvist, and Vissing-Jorgensen (2008) provide direct empirical evidence showing that both the probability of raising a follow-on fund and the size of the follow-on fund increase in the return that investors earned in the previous fund of the same VC, thus suggesting that more successful VCs are able to raise larger follow-on investments.[22] For each VC-backed firm in our sample, we then calculate the average reputation of the VC syndicate that provides the first round of VC financing to that firm. *High Reputation* corresponds to the average market share of funds raised by the VC syndicate being above the sample median, while *Low Reputation* corresponds

[22] Our reputation measure based on the market share of the amount of funds raised by a VC is also quite persistent over time. This is consistent with the evidence in Kaplan and Schoar (2005) and Hochberg, Ljungqvist, and Vissing-Jorgensen (2008) showing persistence in fund performance.

to the average market share of funds raised being below the sample median.[23] For robustness, we also use alternative measures of VC reputation based on the market share of the total amount of investment made by the VC firm and the market share of the market capitalization of IPO exits (on the first trading day) of the VC's portfolio firms (relative to all VC-backed IPOs) over a five-year rolling window (see Ivanov, Krishnan, Masulis, and Singh 2010 and Nahata 2008 for two papers using the IPO-based measure).[24]

In addition to the firm-specific and industry-wide controls mentioned above, we also use several variables as instruments in our switching regression analysis (which we discuss in detail below). The set of instruments we use to capture the potential demand for VC funds from entrepreneurial firms relates to the increase in National Science Foundation (NSF) grants (in real terms) to applied and basic research.[25] Our intuition is that an increase in *NSF Research Grants* could lead to an increase in the establishment of new entrepreneurial firms that may then potentially seek VC investment. To identify the potential supply of VC funds, we use the number of limited partners (LPs), defined as the total number of limited partners investing in VC funds that exist over the previous five years. Using a similar logic, we also use the fraction of the number of LPs in the past five years that are located in the state of the company to capture the potential heterogeneity in the supply of entrepreneurial funds. The greater the availability of LPs and the more of the LPs that are in geographic proximity to the company, the greater should be the ability of local VCs to raise financing, and thus, the greater should be the likelihood of VC backing. This is somewhat similar to the instrument used in Samila and Sorenson (2010, 2011), who use endowment returns of colleges and universities in a metropolitan statistical area (MSA) as their instrument for VC investment in that MSA. In addition, the *AAA Spread*, which is the spread of AAA bonds over five-year Treasury bonds, captures the investment alternatives available to investors that may invest in VC funds. An increase in the spread may lead to a decline in commitments to VC funds, thus lowering overall VC investments. We discuss the significance of using these instruments for our analysis later on in the article.

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23 Such a market-share-based measure may also reflect VC expertise or VC experience in addition to prior success of the VC as measured by internal rates of return. For the purposes of this article, we use the words *reputation* (which is built on prior success) and *experience* or *expertise* interchangeably.

24 We also recalculate our reputation measures by allowing the market shares to be calculated over the entire life of the VC firm as opposed to the last five years. All our results remain qualitatively unchanged to these alternative definitions of VC reputation.

25 *NSF Basic Research Grants* are defined as "systematic study directed toward fuller knowledge or understanding of the fundamental aspects of phenomena and of observable facts without specific applications toward processes or products in mind," while *NSF Applied Research Grants* are defined as "systematic study to gain knowledge or understanding necessary to determine the means by which a recognized and specific need may be met." Further details can be found on the NSF Website at: http://www.nsf.gov/statistics/nsf07303/content.cfm?pub_id=3734&id=1.

## 3. Do Venture Capitalists Improve Firm Efficiency?

### 3.1 Descriptive statistics

As mentioned earlier, the sample of VC-backed firms used in this study comprises all private firms in the LRD that received VC funding between the years 1972 and 2000. In order to benchmark the effect of VC financing properly, we also include in our sample all private firms in the LRD that did not receive VC financing. Panel A of Table 2 presents the summary statistics (means and quasi-medians) of firm characteristics for both VC-backed and non-VC-backed private firms in the LRD during our sample period.[26] We find that VC-backed firms in our sample are on average larger than non-VC-backed firms, in terms of asset value, sales, and total employment. In terms of asset value, VC-backed firms are on average seven times larger than non-VC-backed firms, while the average sales of VC-backed firms is about six times larger than that of non-VC-backed firms. Total cost of materials and total salaries and wages for VC-backed firms are also larger (on average about 5.5 times) than those of non-VC-backed firms, consistent with the argument made by Puri and Zarutskie (2009) regarding the importance of scale in VC financing. In addition, as suggested by anecdotal evidence and several prior papers, we also find that a greater proportion of high-tech firms are VC financed. We find that VCs on average invest in industries that have a higher volatility of firm sales over the past five years, suggesting that VCs tend to invest more in industries that are inherently riskier, and thus the potential contribution that the VC can make to the ultimate success of firms in such an industry is also significantly greater. Finally, we show that the median age at which firms receive their first round of VC financing is three years.

In order to address the potentially unbalanced nature of the LRD sample, we present further summary statistics for our VC-backed sample of firms in Panel B of Table 2. Specifically, we look at VC-backed firms that received their first round of VC financing in census years (i.e., in the years 1972, 1977, 1982, 1987, 1992, and 1997) and compare them with VC-backed firms that received their first round of financing in non-census years. The intuition for this test is as follows: Since the entire universe of manufacturing plants is included in census years, the average VC-backed firm in these years would be significantly smaller in size if indeed there is a significant discrepancy in the coverage of smaller VC-backed firms in non-census years. If it were so, one could argue that we are unable to identify smaller VC-backed firms in non-census years. The results of Panel B, however, rule out this concern, since we find that firms that receive VC financing in census years are not smaller than firms that receive VC financing in non-census years. We use various measures

[26] In order to comply with the confidentiality criteria of the U.S. Census Bureau, we are unable to report the medians of firm characteristics. Therefore, to circumvent this problem, we report quasi-medians, which are the average of the 43rd and the 57th percentiles of each variable and closely approximates the true median value of the variables.

for size such as total assets, total sales, total employment, total salaries and wages, market share, etc., and do not find any statistically significant difference between the size of firms receiving VC financing in census years and the size of firms receiving VC financing in non-census years. This result thus provides

**Table 2**
**Summary statistics of VC- and non-VC-backed firms**

*Panel A: Venture- and Non-Venture-Backed Firms from LRD*

| | | Venture-Backed Firms | Non-Venture-Backed Firms | Difference |
|---|---|---|---|---|
| Total Assets | Mean | 8109 | 1069.45 | 7039.55*** |
| | Quasi-Median | 1687.59 | 195.1 | 1492.49*** |
| | Observations | 1881 | 50114 | |
| Total Sales | Mean | 32730.22 | 5531.58 | 27198.64*** |
| | Quasi-Median | 10976.41 | 1926.3 | 9050.11*** |
| | Observations | 1881 | 50114 | |
| Total Employment | Mean | 250.64 | 53.08 | 197.57*** |
| | Quasi-Median | 106 | 26 | 80*** |
| | Observations | 1881 | 50114 | |
| Materials Cost | Mean | 15844.71 | 2816.96 | 13027.75*** |
| | Quasi-Median | 4688 | 755 | 3933*** |
| | Observations | 1881 | 50114 | |
| Salaries and Wages | Mean | 6306.24 | 1171.83 | 5134.41*** |
| | Quasi-Median | 3054 | 541.5 | 2512.5*** |
| | Observations | 1881 | 50114 | |
| Firm Age at VC Financing | Mean | 8.78 | | |
| | Quasi-Median | 3 | | |
| | Observations | 1881 | | |
| High-Tech Firm | Mean | 0.36 | 0.06 | 0.30*** |
| | Observations | 1881 | 50114 | |
| Industry Risk | Mean | 3768.82 | 1166.06 | 2602.76*** |
| | Quasi-Median | 1925.14 | 720.16 | 1204.99*** |
| | Observations | 1881 | 50114 | |
| Firm Market Share | Mean | 0.01 | 0.00 | 0.01*** |
| | Quasi-Median | 0.00045 | 0.00015 | 0.0003*** |
| | Observations | 1881 | 50114 | |

*Panel B: Venture-Backed Firms Receiving the First Round of VC Financing in Census and Non-Census Years*

| | | Census Year | Non-Census Year | Difference |
|---|---|---|---|---|
| Total Assets | Mean | 8234.00 | 8080.12 | 153.88 |
| | Quasi-Median | 1969.06 | 1587.88 | 381.18 |
| Total Sales | Mean | 33701.59 | 32505.82 | 1195.77 |
| | Quasi-Median | 12219.04 | 10634.01 | 1585.04 |
| Total Employment | Mean | 259.58 | 248.58 | 11.01 |
| | Quasi-Median | 113.50 | 105.00 | 8.50 |
| Material Costs | Mean | 16415.86 | 15712.76 | 703.10 |
| | Quasi-Median | 5545.00 | 4505.50 | 1039.50 |
| Salaries and Wages | Mean | 6424.42 | 6278.94 | 145.48 |
| | Quasi-Median | 3277.50 | 2958.00 | 319.50 |
| Firm Age at VC Financing | Mean | 7.798 | 9.005 | −1.207 |
| | Quasi-Median | 2 | 2.50 | −0.50* |

(*continued*)

**Table 2**
**Continued**

| | | | | |
|---|---|---|---|---|
| High-Tech Firm | Mean | 0.38 | 0.36 | 0.03 |
| Industry Risk | Mean | 3849.20 | 3750.31 | 98.90 |
| | Quasi-Median | 1870.52 | 1946.87 | −76.35 |
| Market Share | Mean | 0.01 | 0.01 | 0.00 |
| | Quasi-Median | 0.00043 | 0.00046 | −0.00003 |

This table reports summary statistics for the sample of VC and similar birth cohort non-VC-backed firms from the manufacturing sector (SIC 2000-3999) in the LRD between 1972 and 2000. *Total Assets* (in thousands of dollars) is constructed via the perpetual inventory method and is the sum of building assets *plus* machinery assets. Total Sales is the total value of shipments in thousands of dollars. *Materials Cost* is the expenses for the cost of materials and parts purchased, resales, contract work, and fuel and energy purchased, in thousands of dollars. *Salaries and Wages* is the sum of total salaries and wages of the firm in thousands of dollars. *Firm Age at VC Financing* is the number of years since the firm first appeared in the LRD sample to the year of first round of VC financing. *High-Tech Firm* is the percentage of firms in the sample that are high-tech companies (i.e., belonging to three-digit SIC codes 357, 366, 367, 372, 381, 382, or 384). *Herfindahl Index* is a measure of concentration of the firm's three-digit SIC industry. *Industry Risk* is the standard deviation of total value of shipments calculated over the prior five-year period in the same three-digit SIC industry as the sample firm. *Firm Market Share* is the firm's market share in terms of sales in the same three-digit SIC industry. To comply with the U.S. Census Bureau confidentiality requirements, we report *Quasi-Medians,* which are the average of the forty-third and the fifty-seventh percentile for each variable. All observations are at the firm-year level. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively. Statistical significance for differences of means and quasi-medians differences correspond to $t$-tests and rank-sum tests, respectively. Panel A reports statistics for VC- versus non-VC-backed firms. Panel B reports statistics in the year of receiving VC financing for VC-backed firms that received VC financing in census years (1972, 1977, 1982, 1987, 1992, 1997) compared with those that received VC financing in non-census years. Variables are winsorized at the 1% and 99% levels.

support to the fact that our analysis uses a representative sample of VC-backed firms within the manufacturing sector and that our results are not driven by the presence of larger VC-backed firms in our sample.

### 3.2 The impact of venture capital financing on firm TFP: Differences in TFP dynamics around the first year of VC financing

In our subsequent analysis, we use TFP as a comprehensive index of firm efficiency. First, we analyze the productivity of VC-backed firms relative to that of non-VC-backed firms. Second, we document the dynamic pattern of productivity changes from *before* to *after* the first round of VC financing, benchmarked against firms not receiving VC financing, and attempt to disentangle the impact on TFP arising due to VC *screening* prior to funding from that arising due to *monitoring* activities of VCs subsequent to funding.[27] We employ a regression framework to analyze these effects, where we include firm and year fixed effects that allow us to precisely control for differences

27 It should be noted that it is not possible for us to differentiate between the effect of monitoring and contracting on firm TFP. Thus, in this article we combine these two effects and for simplicity refer to it as monitoring. It can be argued that the level of monitoring and the stringency of the financial contract are simultaneously determined, since the VC can trade off one for the other. Ultimately, what is important for our analysis is the relative improvement in efficiency that VC firms achieve over non-VC firms subsequent to receiving VC financing and the fact that this improvement in performance and efficiency can be attributed to the involvement of the VC with these firms.

between firms and across time, respectively. Firm fixed effects control for unobserved heterogeneity when this heterogeneity is constant over time and essentially removes this effect by demeaning the regression variables, thereby removing any time-invariant firm-specific unobservable characteristics that may affect the dependent and independent variables in the regression (see, e.g., Wooldridge 2002, pp. 254, 265–74). As a result, while the coefficients in an ordinary least-squares (OLS) regression without firm fixed effects would include the effects of unobserved time-invariant firm characteristics, coefficients in a fixed effects regression would exclude the effect of these characteristics, thus providing a cleaner measure of how VC backing affects the variables under study. As VC financing of firms is distributed over time, we allow for the staggering of the events in our regressions. We also control for time-varying observables of the firm and industry. The methodology adopted in our regression framework throughout this article is consistent with that suggested by Petersen (2009), who advocates using fixed effects and adjusting the standard errors for correlations within clusters. In all regressions, we report standard errors clustered at the firm level. We implement this approach through the following regressions:

$$Y_{it} = \alpha_t + \beta_i + \gamma X_{it} + \delta VCAfter_{it} + \varepsilon_{it} \tag{2}$$

$$\begin{aligned} Y_{it} = \alpha_t + \beta_i + \gamma X_{it} + \delta_1 VCBefore_{-4,0} + \delta_2 VCAfter_{1,4} \\ + \delta_3 VCAfter_{\geq 5} + \varepsilon_{it} \end{aligned} \tag{3}$$

$$Y_{it} = \alpha_t + \beta_i + \gamma X_{it} + \sum_{s=0}^{-4} \delta_1^s VCBefore_{it}^s + \sum_{s=1}^{\geq 5} \delta_2^s VCAfter_{it}^s + \varepsilon_{it}, \tag{4}$$

where $Y_{it}$ is our variable of interest (i.e., firm TFP); $X_{it}$ is a control for firm size and the industry Herfindahl index, which are time varying; $VCAfter_{it}$ in (2) is a dummy variable, which equals one if the firm received VC financing and the observation is in a year after the first round of financing and zero if it either is a firm that did not receive VC financing or is a VC-backed firm, but with the observation belonging to a year prior to the first round of VC financing.[28] In (3), we introduce $VCBefore_{-4,0}$, which is a dummy variable that equals one if the firm received VC financing and the observation is within five years prior to the first round of financing and zero otherwise. Conceptually, this variable is similar to the $VCAfter_{it}$ variable and captures the difference in the TFP between VC-backed and non-VC-backed firms in the years prior to receiving VC financing. We also decompose the $VCAfter_{it}$ variable into

[28] This variable is conceptually similar to the interaction of two dummy variables: $VC * After$, where $VC$ is a dummy variable that equals one if the firm receives VC financing and zero otherwise, and *After* is a dummy variable that equals one if the observation is in a year following the first round of VC financing and zero otherwise. Note that *After* is always zero for non-VC-backed firms. Thus, this specification implicitly takes all firms that have not received VC financing prior to time $t$ as the control group.

two parts: $VCAfter_{1,4}$ captures the changes from years 1 to 4 subsequent to the first round of financing, and $VCAfter_{\geq 5}$ captures the effect on TFP from the fifth year after the first round of financing until exit. This allows us to address how the changes brought about by the VC financing are distributed over time around the first year of receiving VC financing. In order to shed more light on the year-by-year changes in firm TFP due to VC financing, we estimate (4), where we decompose both the $VCBefore_{it}$ and $VCAfter_{it}$ dummies separately for each year. For example, $VCAfter_{it}^{s}$ equals one if the firm receives VC financing and the observation is $s$ years after the first round of financing, where $s = 1, 2, 3, 4$, and $\geq 5$.[29] The dynamic pattern of the effect of VC financing on TFP is captured by the coefficients $\delta' s$ in all three equations. In all specifications, $i$ indexes firms, $t$ indexes years, $\beta_i$ are firm fixed effects, and $\alpha_t$ are year fixed effects. The above specifications are estimated on the entire panel of private firms in the LRD including firms that received VC funding and those that did not. It should, however, be noted that the above fixed effects methodology may not properly capture the screening and monitoring effects of VC backing, if VCs have private information about the firms they invest in and particularly if such information is correlated with future TFP increases. We therefore use two alternative methodologies later on to analyze the causal effect of VC financing on TFP growth (due to the monitoring activities of VCs).

Panel A of Table 3 presents the results showing the effect of VC monitoring and screening on firm TFP. Our estimate of the effect of VC monitoring on a firm's TFP is captured by the $\delta_2' s$ (the coefficients on *VCAfter*), and the effect of VC screening on a firm's TFP is captured by the $\delta_1' s$ (the coefficients on *VCBefore*).[30] As can be seen from Table 3, VCs actively engage in both screening and monitoring. *Regression 2* and *Regression 3* in Panel A of Table 3 show that, on average, firms financed by VCs have 7% higher TFP over the five years prior to receiving funding, compared with firms that do not obtain VC financing, indicating that VCs actively screen firms prior to funding and select the ones with higher levels of productivity. Furthermore, subsequent to funding, TFP of VC-backed firms improves even further, on average to 12% over the next four years, over and above that of non-VC-backed firms, suggesting that VCs actively monitor their investments and improve the performance and efficiency of firms they fund. Finally,

[29] The $VCBefore_{it}^{s}$ dummy is similarly defined. Specifically, $VCBefore_{it}^{0}$ refers to a firm that received VC backing with the observation in the year it received the first round of VC financing, and $VCBefore_{it}^{1}$ refers to a VC-backed firm one year before receiving the first round of financing, and so on.

[30] These results give us an indication of how screening and monitoring activities of VCs impact firm performance and efficiency as measured by TFP. However, these coefficients could potentially be confounded due to an endogeneity problem that arises between VC financing and changes in firm TFP due to selection. We explore this in more detail later and employ an endogenous switching regression methodology as well as a regression discontinuity methodology to accurately capture the relative magnitudes of the impact of screening and monitoring on firm TFP. The qualitative results obtained from this table remain unchanged even after correcting the potential endogeneity issues.

*Regression 4* decomposes the screening and monitoring effects dynamically for every year around the first round of VC funding. As can be seen from the results, for every year prior to receiving funding, VC-backed firms outperform non-VC-backed firms. While there is no apparent trend over the years prior to receiving funding, on average these firms had higher TFP of around 7% relative to non-VC firms, similar to the earlier results. After receiving funding, we observe an increasing trend of TFP for VC-backed firms for five years after the first round of financing, suggesting that the involvement of VCs and their monitoring of these firms leads to an increase in the TFP of these firms over and above that of non-VC-backed firms. This increase is more pronounced after



**Table 3**
**TFP dynamics around year of VC financing**

*Panel A: Regression Results for TFP Changes over Time*

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| VCAfter | 0.121*** | | | | | | |
| | [0.030] | | | | | | |
| VCBefore(−4,0) | | 0.068*** | 0.068*** | | 0.067*** | 0.067*** | 0.069*** |
| | | [0.021] | [0.021] | | [0.021] | [0.021] | [0.021] |
| VCAfter(−1,4) | | 0.120*** | | | 0.118*** | 0.118*** | 0.122*** |
| | | [0.032] | | | [0.032] | [0.032] | [0.032] |
| VCBefore(−4) | | | | 0.042* | | | |
| | | | | [0.022] | | | |
| VCBefore(−3) | | | | 0.088*** | | | |
| | | | | [0.025] | | | |
| VCBefore(−2) | | | | 0.086*** | | | |
| | | | | [0.026] | | | |
| VCBefore(−1) | | | | 0.053* | | | |
| | | | | [0.027] | | | |
| VCBefore(0) | | | | 0.071** | | | |
| | | | | [0.030] | | | |
| VCAfter(1) | | | 0.108*** | 0.108*** | | | |
| | | | [0.034] | [0.035] | | | |
| VCAfter(2) | | | 0.106*** | 0.106*** | | | |
| | | | [0.035] | [0.035] | | | |
| VCAfter(3) | | | 0.13*** | 0.13*** | | | |
| | | | [0.035] | [0.035] | | | |
| VCAfter(4) | | | 0.136*** | 0.137*** | | | |
| | | | [0.039] | [0.039] | | | |
| VCAfter(≥ 5) | | 0.188*** | 0.189*** | 0.189*** | 0.186*** | 0.185*** | 0.190*** |
| | | [0.042] | [0.042] | [0.043] | [0.042] | [0.042] | [0.042] |
| Firm Age | | | | | −0.006* | 0.001 | |
| | | | | | [0.003] | [0.006] | |
| Firm Age Squared | | | | | | −0.005 | |
| | | | | | | [0.232] | |
| Firm Size | −0.057*** | −0.057*** | −0.057*** | −0.057*** | −0.056*** | −0.056*** | −0.057*** |
| | [0.003] | [0.003] | [0.003] | [0.003] | [0.003] | [0.003] | [0.003] |
| Herfindahl Index | −0.019 | −0.018 | −0.018 | −0.018 | −0.019 | −0.019 | −0.018 |
| | [0.020] | [0.020] | [0.020] | [0.020] | [0.020] | [0.020] | [0.020] |
| Firm Fixed Effects | Y | Y | Y | Y | Y | Y | Y |
| Year Fixed Effects | Y | Y | Y | Y | Y | Y | Y |
| Firm Age Fixed Effects | N | N | N | N | N | N | Y |
| Observations | 524806 | 524806 | 524806 | 524806 | 524806 | 524806 | 524806 |
| Adj. $R^2$ | 0.414 | 0.414 | 0.414 | 0.414 | 0.414 | 0.414 | 0.414 |

(*continued*)

**Table 3**
**Continued**

*Panel B: TFP Change over Time*

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| VCAfter(1,4) – VCBefore(4,0) | | 0.052** | | | 0.051** | 0.051** | 0.052** |
| | | [0.021] | | | [0.021] | [0.021] | [0.021] |
| VCAfter(≥5) – VCBefore(4,0) | | 0.119*** | | | 0.119*** | 0.118*** | 0.121*** |
| | | [0.033] | | | [0.033] | [0.033] | [0.033] |
| VCAfter(1) – VCBefore(4,0) | | | 0.040* | | | | |
| | | | [0.024] | | | | |
| VCAfter(2) – VCBefore(4,0) | | | 0.038 | | | | |
| | | | [0.026] | | | | |
| VCAfter(3) – VCBefore(4,0) | | | 0.062** | | | | |
| | | | [0.026] | | | | |
| VCAfter(4) – VCBefore(4,0) | | | 0.069** | | | | |
| | | | [0.029] | | | | |
| VCAfter(≥5) – VCBefore(4,0) | | | 0.121*** | | | | |
| | | | [0.033] | | | | |
| VCAfter(1) – VCBefore(1) | | | | 0.056* | | | |
| | | | | [0.029] | | | |
| VCAfter(2) – VCBefore(1) | | | | 0.054* | | | |
| | | | | [0.029] | | | |
| VCAfter(3) – VCBefore(1) | | | | 0.077*** | | | |
| | | | | [0.030] | | | |
| VCAfter(4) – VCBefore(1) | | | | 0.084*** | | | |
| | | | | [0.032] | | | |
| VCAfter(≥5) – VCBefore(1) | | | | 0.136*** | | | |
| | | | | [0.036] | | | |

This table reports results for panel data regressions in which the dependent variable is the TFP of a firm for a given year. The independent variables in Panel A are *VCAfter,* which is a dummy variable that equals one for years after the firm gets the first round of VC financing and zero otherwise; *VCBefore(–4,0),* which is a dummy variable that equals one for years –4 to 0 prior to obtaining the first round of VC financing and zero otherwise; *VCAfter(1,4),* which is a dummy variable that equals one for years 1 to 4 after obtaining the first round of VC financing and zero otherwise; *VCBefore(–t)* for all $0 \leq t \leq 4$, which equals one in year *t* before VC financing and zero otherwise; *VCAfter(t)* for all $0 \leq t \leq 4$, which equals one in year *t* after VC financing and zero otherwise; *VCAfter*(≥5), which equals one in or after year 5 of VC financing and zero otherwise; *Firm Size,* which is the natural log of the firm's capital stock in a given year; *Herfindahl Index,* which is the concentration of the firm's three-digit SIC industry; *Firm Age*, which is the natural logarithm of the number of years since the firm first appeared in the LRD sample; *Firm Age Squared*, which is the square of the log of age; and year, firm, and age fixed effects. Panel A reports the regression coefficient estimates and their statistical significances. Panel B reports TFP changes over various time periods and their statistical significances. Heteroscedasticity corrected robust standard errors, which are clustered on firms, are in brackets. All regressions are estimated with an intercept term. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

5 years (and beyond) of receiving funding when the TFP for VC-backed firms increases to about 19% above that of non-VC-backed firms. In most cases, the coefficients of interest in *Regression 1* to *Regression 4* are significant at the 1% level. In *Regressions 5, 6,* and *7,* we control for log of firm age, log of firm age squared, and age dummies, respectively. We find similar results after controlling for firm age, suggesting that our results are robust to controlling for life-cycle effects.

Panel B of Table 3 presents the net effect on firm TFP that can be attributed to the monitoring abilities of VCs. Based on *Regression 2* to *Regression 4*, we find that the average increase in TFP from before receiving VC financing to four years after receiving VC financing is around 5% for VC-backed firms.

This increase is around 12% for VC-backed firms when we compute the difference in TFP after year 5. In *Regression 4*, we compute the net effect of monitoring year by year for VC-backed firms relative to the year prior to receiving financing. Consistent with the evidence presented so far, we observe that there is a monotonically increasing effect of VC monitoring on TFP. In the two years after the first financing round, the impact of monitoring is around 5.5%, which monotonically increases to around 14% after the fifth year subsequent to the first round of financing.[31] Overall, the results from Panels A and B of Table 3 suggest that VCs actively engage in both screening and monitoring. On average, we find that due to their screening ability, VCs invest in firms that have around 7% higher TFP than firms that do not receive VC funding. We also find that the net effect of monitoring on the TFP of firms is between 5.5% and around 14%. Thus, our results suggest that the impact of screening and monitoring of VCs on firm performance and efficiency are both important and on average they have similar magnitudes, if we consider solely the net effect of monitoring during the first two years after receiving the first round of financing. Our results on the change in TFP from before to after VC financing are also robust to age or life-cyle effects, which we control for in *Regressions 5*, *6*, and *7*. Overall, the results presented here are economically significant.[32]

As a robustness check for our results, we use alternative measures of performance. We use value added per worker or labor productivity, which is defined as total sales divided by the number of workers. In addition, we also use ROA, defined as total sales minus production costs divided by firm capital stock, as another alternative measure of performance to check our results. These measures do not have the desirable theoretical properties of TFP, but they do have familiar statistical properties, since they are not computed from a regression. We find qualitatively similar results using labor productivity and ROA to those using TFP.[33] In our analysis, failed firms will drop out of the

[31] It should be noted, however, that this entire net effect of monitoring on firm TFP should be attributed not only to the first round of VC financing. Since there are typically multiple rounds of financing that a VC-backed firm receives, the net impact of monitoring that we find here could potentially be attributed to multiple rounds of financing. As mentioned earlier, our TFP measure is independent of the scale of production, so while this net increase in TFP is not due to the direct effect of capital infusion of the later rounds, it could potentially be argued that the level of monitoring of a VC could increase with the amount of investment made in the firm. Thus, this could lead us to observe the net higher increase in TFP for the years that are further away from the initial year of investment. We analyze this in Section 3.5, when we look at TFP improvements around subsequent rounds of VC financing.

[32] A 10% increase in TFP is equivalent to a 10% increase in revenue, holding all input costs constant. As discussed in Section 2.2, Schoar (2002) suggests a methodology to relate TFP to profits as well. The higher TFP of 7% in VC-backed firms due to screening, and the increase in net average TFP of 12% due to monitoring of VCs, translate to an increase in profits of approximately 24.5% and 42%, respectively. These increases in profits are calculated based on the assumption of a revenue margin of 40% over costs, *ceteris paribus*. Note here that the elasticity of profits to productivity is greater than one and the smaller the profit margin, the higher the elasticity of profits to productivity. This means that the 40% profit margin used in the above calculation is a conservative number, since a lower profit margin will actually increase the effect of TFP change on profits.

[33] Since the VentureXpert data are noisier prior to 1980, we redo our analysis using data for the subset of VC-backed firms that receive VC financing after 1980. Our results are qualitatively similar to those reported here.

sample. However, any survivorship bias as a result of this is unlikely to drive our results. This is because even though firms that fail drop out of our sample, we find that non-VC-backed firms are more likely to fail than VC-backed firms. Thus, any survivorship bias would make the performance of non-VC-backed firms appear better, thus biasing the analysis against our findings.[34] To further alleviate concerns that our results are driven by potential survivorship biases, we remove all firms that fail within our sample period (both VC- and non-VC-backed), and repeat our TFP analysis on this restricted sample. In other words, we run our analysis with firms that survive throughout the sample period. Our results are qualitatively similar to the ones reported here when we use this restricted sample.

The above results are consistent with prior articles in the literature that argue that VCs create value through screening and monitoring, such as Lerner (1995) and Hellmann and Puri (2000, 2002). Our results complement these earlier studies and present direct evidence regarding the impact of such activities of VCs on firm performance and efficiency over and above that of non-VC-backed private firms. In particular, this is the first study to directly relate the productivity of VC-backed firms to that of non-VC-backed firms both prior to and after receiving VC financing, thus quantifying the impact on firm TFP due to VC involvement.

### 3.3 TFP dynamics for early- and late-stage venture capital-backed firms

A potential concern with our analysis is that our sample of VC-backed manufacturing firms may be biased toward later-stage VC-backed firms, and thus our results may not be generalizable to early-stage VC-backed firms.[35] To address this concern, we run our fixed effect regressions separately on the set of early- and late-stage VC-backed firms. Early-stage firms are firms whose stage at the first round of VC financing is defined in SDC VentureXpert as “Early stage” or “Seed/Startup.” Table 4 reports the results of the fixed effects regressions for the set of early- and late-stage VC-backed firms. We find that the results for early- and late-stage VC-backed firms are similar to the regression results for the overall sample. Further, the monitoring impact of venture capitalists is stronger for early-stage VC-backed firms: The increase in TFP over the four years after receiving VC financing from the five years prior to receiving VC financing is statistically significant only for the sample of early-stage VC-backed firms. The economic magnitude of TFP increases from before to after VC financing is greater for early-stage VC-backed firms compared with those for late-stage VC-backed firms.

34 Puri and Zarutskie (2009) also find that, in their sample, VC-backed firms are less likely to fail than non-VC-backed firms, on average.

35 Approximately 42% of our sample consists of early-stage VC-backed firms.

**Table 4**
**TFP dynamics around year of VC financing for early- and late-stage firms**

*Panel A: Regression Results for TFP Changes over Time*

| | Early-stage | Late-Stage |
|---|---|---|
| VCBefore(–4,0) | 0.067* | 0.08*** |
| | [0.039] | [0.023] |
| VCAfter(1,4) | 0.201*** | 0.108*** |
| | [0.069] | [0.028] |
| VCAfter(≥5) | 0.313*** | 0.16*** |
| | [0.095] | [0.03] |
| Firm Size | −0.064*** | −0.065*** |
| | [0.003] | [0.002] |
| Herfindahl Index | −0.013 | −0.016 |
| | [0.017] | [0.016] |
| Year Fixed Effects | Y | Y |
| Firm Fixed Effects | Y | Y |
| Observations | 515581 | 520728 |
| Adj. $R^2$ | 0.423 | 0.423 |

*Panel B: TFP Changes over Time*

| | | |
|---|---|---|
| VCAfter(1,4) – VCBefore(4,0) | 0.133*** | 0.028 |
| | [0.466] | [0.018] |
| VCAfter(≥5) – VCBefore(4,0) | 0.246*** | 0.080*** |
| | [0.073] | [0.029] |

This table reports results for panel data regressions where the dependent variable is the TFP of a firm for a given year. The regression results are reported for subsamples of early- and late-stage firms, where an early-stage firm is a firm that receives first round of VC financing at the "Early" or "Startup/seed" stage, based on VentureXpert classification. The independent variables in Panel A are *VCBefore(–4,0),* which is a dummy variable that equals one for years –4 to 0 prior to obtaining the first round of VC financing and zero otherwise; *VCAfter(1,4)*, which is a dummy variable that equals one for years 1 to 4 after obtaining the first round of VC financing and zero otherwise; *VCAfter*(≥5), which equals one in or after year 5 of VC financing and zero otherwise; *Firm Size*, which is the natural log of the firm's capital stock in a given year; *Herfindahl Index,* which is the concentration of the firm's three-digit SIC industry; and firm and year fixed effects. Panel A reports the regression coefficient estimates and their statistical significances. Panel B reports TFP changes over various time periods and their statistical significances. Heteroscedasticity corrected robust standard errors, which are clustered on firms, are in brackets. All regressions are estimated with an intercept term. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

These results are consistent with our overall results in the previous section, and thus mitigate concerns that our results are driven by late-stage VC-backed firms. Rather, the economic effect of VC monitoring on firm TFP is greater in the sample of early-stage VC-backed firms.[36]

### 3.4 The effect of option compensation on TFP dynamics

Hellmann and Puri (2002) argue that VCs introduce option-based compensation in the firms that they back. Thus, a potential concern with our TFP results above is that our TFP calculations understate the magnitude of option compensation, which in turn might mechanically bias our TFP measure

[36] We also repeat this analysis by splitting our sample into young and old VC-backed firms, based on firm age at VC financing. Our results are similar to the ones reported in the article: We find that TFP is greater prior to VC financing and increases even more subsequent to VC financing for both younger and older firms. The monitoring effect, however, is stronger for younger VC-backed firms.

upward, particularly for VC-backed firms. To alleviate this concern, we perform an adjustment to our TFP measure.[37] Using aggregate industry-level option compensation data, we adjust the labor input in the Cobb-Douglas production function to reflect the greater expense of providing option compensation. In particular, we adjust the salaries and wages of white-collar employees in VC-backed firms, starting from the year in which they receive VC financing, upward by a factor that is equal to one plus the ratio of average options to cash compensation in the year and three-digit SIC industry of the VC-backed firm. We do not adjust the labor input for the non-VC-backed firms or for VC-backed firms in the years prior to receiving VC financing, thus biasing the analysis *against* finding higher TFP growth for VC-backed firms. We then run our TFP calculation regressions, described in Section 2.1, using this adjusted labor input measure and calculate firm-level TFP as before.

We obtain data on the stock options and compensation from the Execucomp database. Since this database only provides data from 1992 onward, we make two different assumptions about the option adjustment prior to 1992. In the first adjustment, we fix option compensation levels prior to 1992 at the 1992 level for each three-digit SIC industry. This is a conservative adjustment (since it lowers the TFP much more for VC-backed firms) because prior literature (e.g., Frydman and Saks 2010) suggests that option compensation levels had been steadily increasing over the 1980s and the 1990s. In the second adjustment, we calculate the average annual compounded growth rate of options in the years after 1992 for each three-digit SIC industry, and use this growth rate to decrease the option compensation adjustment backward from 1992.[38] We also repeat this adjustment and TFP calculation using combined stock and option compensation.

Our results using the above adjusted TFP measures are reported in Table 5. They indicate that even after adjusting for option compensation, our results remain similar to the ones reported earlier. In particular, we find that VC-backed firms have higher TFP than non-VC-backed firms prior to the VC financing, and their option-adjusted TFP increases subsequent to investment by the VC. Our results hold for both types of pre-1992 option adjustment methods.[39] Further, our results remain similar when we adjust for combined stock and option compensation. Thus, the results in this section mitigate concerns that our results may be driven by overstatement of the TFP due to not adjusting our labor measure appropriately for the introduction of option compensation in VC-backed firms.

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[37] We are grateful to an anonymous referee for the suggestion to adjust our labor measure for option compensation.

[38] In particular, we calculate the average growth rate for each three-digit SIC industry as $g = ((1+g_1)(1+g_2)\ldots(1+g_n))^{1/n} - 1$. We then adjust options to cash compensation ratio backward as $opt/comp_{t-1} = (opt/comp_t)/(1+g)$.

[39] Our results are also qualitatively similar when we use data only after 1992 for the fixed effects regressions.

**Table 5**
**Option- and stock-adjusted TFP dynamics around year of VC financing**

*Panel A: Regression Results for TFP Changes over Time*

| | Option adjusted, Fixed level pre-1992 | Stock and Option adjusted, Fixed level pre-1992 | Option adjusted, Declining level pre-1992 | Stock and Option adjusted, Declining level pre-1992 |
|---|---|---|---|---|
| VCBefore(–4,0) | 0.06*** | 0.06*** | 0.06*** | 0.06*** |
| | [0.019] | [0.019] | [0.019] | [0.019] |
| VCAfter(1,4) | 0.098*** | 0.095*** | 0.100*** | 0.098*** |
| | [0.024] | [0.024] | [0.024] | [0.024] |
| VCAfter(≥5) | 0.118*** | 0.116*** | 0.118*** | 0.115*** |
| | [0.026] | [0.026] | [0.026] | [0.026] |
| Firm Size | −0.062*** | −0.062*** | −0.062*** | −0.062*** |
| | [0.003] | [0.003] | [0.003] | [0.003] |
| Herfindahl Index | −0.034 | −0.034 | −0.033 | −0.033 |
| | [0.023] | [0.023] | [0.023] | [0.023] |
| Year Fixed Effects | Y | Y | Y | Y |
| Firm Fixed Effects | Y | Y | Y | Y |
| Observations | 524402 | 524402 | 524402 | 524402 |
| Adj. $R^2$ | 0.381 | 0.381 | 0.381 | 0.381 |

*Panel B: TFP Changes over Time*

| | | | | |
|---|---|---|---|---|
| VCAfter(1,4) – VCBefore(4,0) | 0.037* | 0.035* | 0.040** | 0.038** |
| | [0.019] | [0.019] | [0.019] | [0.019] |
| VCAfter(≥5) – VCBefore(4,0) | 0.058** | 0.055** | 0.058** | 0.056** |
| | [0.023] | [0.023] | [0.023] | [0.023] |

This table reports results for panel data regressions in which the dependent variable is the TFP of a firm for a given year adjusted for option grants or stock and option grants. The first two columns report the results for TFP measures that are calculated assuming that option and stock grants relative to cash compensation in all years prior to 1992 are fixed at 1992 levels. The second two columns report the results for TFP measures that are calculated assuming that option and stock grants relative to cash compensation in years prior to 1992 decline going back in time at a constant imputed rate. The independent variables in Panel A are *VCBefore(−4,0)*, which is a dummy variable that equals one for years –4 to 0 prior to obtaining the first round of VC financing and zero otherwise; *VCAfter(1,4)*, which is a dummy variable that equals one for years 1 to 4 after obtaining the first round of VC financing and zero otherwise; *VCAfter*(≥5), which equals one in or after year 5 of VC financing and zero otherwise; *Firm Size*, which is the natural log of the firm's capital stock in a given year; *Herfindahl Index*, which is the concentration of the firm's three-digit SIC industry; and firm and year fixed effects. Panel A reports the regression coefficient estimates and their statistical significances. Panel B reports TFP changes over various time periods and their statistical significances. Heteroscedasticity corrected robust standard errors, which are clustered on firms, are in brackets. All regressions are estimated with an intercept term. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

## 3.5 Impact of screening and monitoring: Differences in TFP dynamics around VC financing rounds

In this section, we analyze the above dynamics in TFP changes round-by-round for VC-backed firms to see how the TFP improvements documented above may be spread over different rounds of venture financing received by the firm. We do this by estimating a modified version of (3) where we replace the *VCAfter* dummies with *VCAfter Round* dummies. Specifically, we look at TFP improvements after *Round 1*, *Round 2*, *Round 3*, and *Round 4* or greater. The results are presented in Table 6. The round-by-round results, reported in Panel A of Table 6, are consistent with those discussed in the previous section and show that at each round after receiving financing and in the years prior to

**Table 6**
**TFP dynamics around rounds of VC financing**

*Panel A: Regression Results for TFP Change over Rounds*

| | |
|---|---|
| VCBefore | 0.057*** |
| | [0.020] |
| VCAfter(R1) | 0.102*** |
| | [0.033] |
| VCAfter(R2) | 0.192*** |
| | [0.048] |
| VCAfter(R3) | 0.189*** |
| | [0.060] |
| VCAfter(≥R4) | 0.155*** |
| | [0.046] |
| Firm Size | −0.057*** |
| | [0.003] |
| Herfindahl Index | −0.016 |
| | [0.020] |
| Firm Fixed Effects | Y |
| Year Fixed Effects | Y |
| Observations | 523750 |
| Adj. $R^2$ | 0.414 |

*Panel B: TFP Change over Rounds*

| | |
|---|---|
| VCAfter(R1) – VCBefore | 0.045* |
| | [0.026] |
| VCAfter(R2) – VCBefore | 0.135*** |
| | [0.043] |
| VCAfter(R3) – VCBefore | 0.131** |
| | [0.056] |
| VCAfter(≥R4) – VCBefore | 0.098** |
| | [0.043] |
| VCAfter(R2) – VCAfter(R1) | 0.090** |
| | [0.039] |
| VCAfter(R3) – VCAfter(R2) | −0.004 |
| | [0.060] |
| VCAfter(≥R4) – VCAfter(R3) | −0.03 |
| | [0.056] |

This table reports results for panel data regressions in which the dependent variable is the TFP of a firm for a given year. The independent variables in the regression are *VCBefore*, which is a dummy variable that equals one for all years prior to the first round of VC financing and zero otherwise; *VCAfter(R1)*, which is a dummy variable that equals one for years after the first round but before the second round of VC financing and zero otherwise; *VCAfter(R2)*, which is a dummy variable that equals one for years after the second round but before the third round of VC financing and zero otherwise; *VCAfter(R3)*, which is a dummy variable that equals one for years after the third round but before the fourth round of VC financing and zero otherwise; *VCAfter*(≥*R*4), which is a dummy variable that equals one for years after the fourth round of VC financing and zero otherwise; *Firm Size*, which is the natural log of the firm's capital stock in a given year; *Herfindahl Index*, which is the concentration of the firm's three-digit SIC industry; and firm and year fixed effects. Panel A reports the regression coefficient estimates and their statistical significances. Panel B reports TFP changes over various rounds and their statistical significances. Heteroscedasticity corrected robust standard errors, which are clustered on firms, are in brackets. All regressions are estimated with an intercept term. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

receiving VC financing, the TFP of VC-backed firms is significantly greater than the TFP of non-VC-backed firms.

In Panel B of Table 6, we show that the TFP for VC-backed firms after each round is significantly greater than the TFP of the firms prior to receiving VC financing. Additionally, the TFP in *Round 2* (i.e., between round 2 and

3) is significantly greater than the TFP in *Round 1* (i.e., between round 1 and 2), suggesting a monotonic increase in TFP from round 1 until the end of round 2 for VC-backed firms. While the TFP improvement after *Round 1* and before *Round 2* is around 5%, the TFP improvement after *Round 2* and before *Round 3* is significantly greater at 9%. However, TFP after *Round 3* is not significantly greater than the TFP before that round. This result therefore suggests that increases in TFP for VC-backed firms is achieved within the first couple of rounds of VC involvement with the firm.

## 4. Effect of VC Monitoring on TFP

### 4.1 Switching regressions with endogenous switching

In this section, we present further evidence on the impact of monitoring by VCs on firm TFP and efficiency. Our results from the previous sections suggest that VCs screen firms and thus the VC-firm matching may be nonrandom. For instance, VCs may select higher-quality firms to invest in, leading to higher performance subsequent to VC investment. In other words, this matching may potentially confound the effects of VC monitoring on firm performance with the effects arising due to VC screening. Thus, to isolate the effect of VC monitoring, we are interested in the following "what-if" type of question: What would the performance of a VC-backed firm have been *had it not received* VC financing? Similarly, what would the performance of a non-VC-backed firm have been *had it received* VC financing? The answer to these questions holds the impact of screening on TFP constant and separates out the effect of monitoring on firm TFP due to VC financing.

To implement this, we adopt a two-step Heckman-type estimation structure and employ a switching regression with endogenous switching methodology to disentangle the impact of screening and monitoring on TFP of VC-backed firms. This procedure is discussed in detail in Heckman (1979) and Maddala (1983) and is a generalized version of the traditional Heckman model and therefore accounts for the effect of unobservables (using which VCs may select better-quality firms) by using inverse Mills ratios. In particular, we use a first-stage regression to predict the probability of VC backing and the inverse Mills ratios for VC-backed and non-VC-backed firms. The inverse Mills ratio has been used in the prior literature as a measure of unobserved factors that affect the selection of firms for treatment (which is VC financing in our context) as well as the outcome variable (which is post-VC financing TFP improvement in our context). Thus, the inverse Mills ratio captures unobservable information using which VCs may select better-quality firms. We then regress TFP growth on the inverse Mills ratio and control variables in the second stage of the estimation separately for VC- and non-VC- backed firms. Finally, the predicted values of TFP growth from the second-stage estimates are used to conduct the above-mentioned "what-if" analysis. Thus, the switching regression analysis measures the effect of the treatment (i.e., VC backing) on the outcome variable (i.e., TFP growth) through a "what-if" or counterfactual analysis rather than estimating it from a regression. This model appears in Lee (1978) in his study of unionism and wage rates, Dunbar (1995) in his study of the use of warrants as underwriter compensation, and more recently in Fang (2005) in her study on investment bank reputation and the price and quality of bond underwriting services provided by them. Please see the technical appendix (Appendix A) for a more detailed description of this methodology.

We control for firm-level characteristics that could affect VC-firm matching, such as the average five-year prior TFP, firm size, the number of plants operated by the firm, firm age, market share, and whether the firm operates in a high-tech industry or not; industry-level characteristics such as the industry Herfindahl index and the riskiness of the industry in which the firm operates; marketwide characteristics such as the S&P 500 return; and time dummies capturing the 1980s, the 1990s, and the Internet-bubble period. In addition, we also include several instruments that are correlated with the demand and supply of venture funds in the economy but are independent of the future performance of VC-backed firms—namely, the increase in NSF applied research grants, the increase in NSF basic research grants, the total number of LPs, the fraction of total LPs in the same state as the entrepreneurial firm, and the AAA spread. These instruments provide us with exogenous variation in terms of both the supply of, and the demand for, VC funds that affect the selection equation (i.e., the matching outcome between VCs and entrepreneurial firms) but do not directly affect firm performance.

The results of the switching regression analysis, reported in Panel A of Table 7, show that prior average TFP, firm size, number of plants operated by the firm, Herfindahl index, and the high-tech dummy are all positive and significant determinants of receiving VC financing. Firm age and industry risk are negative and significant determinants of VC financing. With regard to the instruments, we find that the coefficient on the increase in NSF applied research grants is positive and significant, while the coefficient on the increase in NSF basic research grants is negative, suggesting that greater applied research may lead to an increase in entrepreneurial firm formation, thus leading to greater demand for VC funds. Similarly, the coefficients on the number of LPs and the fraction of in-state LPs are also positive and significant, suggesting a greater probability of receiving VC funding in the presence of a greater number of LPs. As expected, the coefficient on AAA spread is negative and significant, suggesting that availability of funds to VCs is an important criterion for VC financing, consistent with Gompers and Lerner (1998). We then augment the second-stage regressions for VC-backed and non-VC-backed firms with the inverse Mills ratio calculated from the first stage to account for endogenous selection (based on unobservable factors).

The second-stage results show that while the inverse Mills ratio is positive and significant for VC-backed firms, it is insignificant for non-VC-backed firms. This suggests that VCs may also use some unobservable factors when

**Table 7**
**Switching regressions with endogenous switching for VC- and non-VC-backed firms**

*Panel A: Switching Regressions with Endogenous Switching*

| First-stage | | Second-stage | | |
|---|---|---|---|---|
| *Dependent Variable: VC Backing Dummy* | | *Dependent Variable: TFP Growth* | | |
| | | | VC-Backed Firms | Non-VC-Backed Firms |
| Number of LPs | 0.0004** | Inverse Mills Ratio | 0.212** | 0.035 |
| | [0.000] | | [0.086] | [0.386] |
| Fraction of Instate LPs | 0.575* | Average 5 Year Prior TFP | −0.779*** | −0.921*** |
| | [0.311] | | [0.062] | [0.009] |
| Increase in NSF Applied Research Grants | 0.445*** | Firm Size | 0.068*** | −0.027*** |
| | [0.127] | | [0.024] | [0.003] |
| Increase in NSF Basic Research Grants | −0.442** | Number of Plants | 0.000 | 0.002** |
| | [0.193] | | [0.001] | [0.001] |
| AAA Spread | −0.115** | Herfindahl Index | −0.086 | −0.031 |
| | [0.048] | | [0.086] | [0.023] |
| Average 5 Year Prior TFP | 0.093** | Firm Age | −0.002 | −0.006*** |
| | [0.043] | | [0.003] | [0.001] |
| Firm Size | 0.212*** | Firm Market Share | −0.066 | 0.214** |
| | [0.012] | | [0.110] | [0.097] |
| Number of Plants | 0.005*** | Industry Risk | −0.001 | 0.0004 |
| | [0.001] | | [0.001] | [0.0003] |
| Herfindahl Index | 0.332* | High-Tech Firm | 0.081 | −0.003 |
| | [0.194] | | [0.05] | [0.023] |
| Firm Age | −0.005** | S&P 500 Returns | −0.009 | 0.003 |
| | [0.002] | | [0.025] | [0.002] |
| Firm Market Share | −0.015 | 1980s Dummy | 0.132*** | 0.008 |
| | [0.36] | | [0.041] | [0.006] |
| Industry Risk | −0.01* | 1990s Dummy | 0.068 | 0.004 |
| | [0.005] | | [0.045] | [0.008] |
| High-Tech Firm | 0.571*** | Bubble Dummy | 0.101* | 0.015 |
| | [0.044] | | [0.056] | [0.011] |
| S&P 500 Returns | 0.078 | | | |
| | [0.133] | | | |

(*continued*)

**Table 7**
**Continued**

*Panel A: Switching Regressions with Endogenous Switching*

| First-stage | | Second-stage | | |
|---|---|---|---|---|
| *Dependent Variable: VC Backing Dummy* | | *Dependent Variable: TFP Growth* | | |
| | | | VC-Backed Firms | Non-VC-Backed Firms |
| 80s Dummy | 0.231*** | | | |
| | [0.087] | | | |
| 90s Dummy | −0.025 | | | |
| | [0.157] | | | |
| Bubble Dummy | 0.296* | | | |
| | [0.152] | | | |
| | | Firm Fixed Effects | Y | Y |
| Observations | 361334 | Observations | 5714 | 292815 |
| *Chi*$^2$ | 1234.9*** | Adj. $R^2$ | 0.665 | 0.69 |

(*continued)*

**Table 7**
**Continued**

*Panel B: Actual and Hypothetical TFP Growth for VC vs. Non-VC-backed Firms*

| | Actual TFP growth for VC-backed firms | TFP growth for VC-backed firms if they had not obtained VC financing | TFP growth improvement |
|---|---|---|---|
| Mean | 0.001 | –0.184 | 0.185***,+++ |
| | Actual TFP growth for non-VC-backed firms | TFP growth for non-VC-backed firms if they had obtained VC financing | TFP growth deterioration |
| Mean | –0.029 | 0.014 | –0.043***,+++ |

This table reports the results from an endogenous switching regression model in which the dependent variable in the first stage is whether or not a firm gets VC financing in a given year (*VC Backing Dummy*) and the associated "what-if" analysis. Panel A reports the regression results. The time series for each firm that gets VC financing terminates in the year of obtaining the first round of VC financing. The independent variables in the regressions are: *Average 5 year prior TFP*, which is the average TFP over the past five years starting from the current year; *Increase in NSF Applied Research Grants*, which is the average change over the past five years in the real National Science Foundation applied research grants ($ million); *Increase in NSF Basic Research Grants*, which is the average change over the past five years in the real National Science Foundation basic research grants ($ million); *AAA Spread*, which is the spread of AAA bonds over five-year treasury bonds in the current year; *Number of LPs*, which equals the total number of limited partners that exist over the previous five years; *Fraction of Instate LPs*, which is the fraction of the number of LPs in the past five years that are located in the state of the company; *Firm Size*, which is the one-year lagged value of natural log of the firm's capital; *Number of Plants*, which is the natural logarithm of the one-year lagged value of the number of plants in the firm; *Herfindahl Index*, which is the one-year lagged value of the measure of concentration of the firm's three-digit SIC industry; *Firm Age*, which is the natural logarithm of the one-year lagged value of the number of years since the firm first appeared in the LRD sample; *Firm Market Share*, which is the one-year lagged value of the firm's market share in terms of sales in the same three-digit SIC industry; *Industry risk*, which is the one-year lagged value of the median standard deviation of the total value of shipments calculated over a prior five-year period for all firms in the same three-digit SIC industry as the sample firm; *High-Tech Firm*, which is a dummy variable that takes the value 0 if the one-year lagged value of the firm's three-digit SIC code is 357, 366, 367, 372, 381, 382, or 384; *1980s Dummy*, which takes the value one for years between 1980 and 1989 and zero otherwise; *1990s Dummy*, which takes the value one for years between 1990 and 1998 and zero otherwise; and *Bubble Dummy*, which takes the value one for years 1999 and 2000 and zero otherwise. The dependent variable in the second-stage regression is the *TFP Growth*, which is defined as the difference between the average TFP over the next five years and the average TFP over the past five years starting from the current year. The independent variables in these regressions are: the *Inverse Mills Ratio* from the first stage and all the independent variables from the first stage except for *Increase in NSF Applied Research Grants, Increase in NSF Basic Research Grants, Number of LPs, Fraction of Instate LPs,* and *AAA Spread.* In the first-stage regressions, heteroscedasticity corrected robust standard errors, which are clustered on firms, are in brackets, while in the second-stage regressions bootstrapped standard errors, which are clustered on firms, are in brackets. All regressions are estimated with an intercept term. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively. Panel B reports the result of a "what-if" analysis based on the results of the switching regression model in Panel A. It reports the actual TFP growth around the first round of VC financing for VC-backed firms, the TFP growth if VC-backed firms did not receive VC financing, and the difference between actual and hypothetical TFP growths (*TFP Growth Improvement*). The table also reports the TFP growth for non-VC-backed firms if they had received VC financing, the actual TFP growth of non-VC-backed firms, and the difference between the actual and hypothetical TFP growths (*TFP Growth Deterioration*). ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively, for *t*-test of mean difference. +++, ++, and + represent statistical significance at the 1%, 5%, and 10% levels, respectively, for the rank-sum test.

they select firms to invest in, and these factors (that they condition the selection on) positively affect the future TFP growth of firms receiving VC financing. Therefore, properly controlling for this effect enables us to attribute the residual TFP growth of VC-backed firms to the monitoring activities of VCs. We find that lower average prior TFP leads to higher future TFP growth in both VC- and non-VC-backed firms. While larger firms have higher future TFP growth among VC-backed firms, smaller firms, multi-plant firms, younger firms, and firms with greater market share have higher future TFP growth among non-VC-backed firms.

Panel B of Table 7 presents the results of our counterfactual analysis of VC-backed versus non-VC-backed firms. We obtain the counterfactual values of TFP growth for VC-backed firms as the predicted values of the non-VC-backed regression and the corresponding inverse Mills ratio (using VC-backed firm data), and vice versa. These results suggest that, on average, VC-backed firms achieve a TFP growth of approximately 18.5% higher than what the *same firms* would have achieved had they not received VC financing, suggesting a monitoring effect. The second row shows a hypothetical improvement in TFP growth of 4.3% for non-VC-backed firms had the *same firms* received VC financing. The magnitude of this improvement attributed to monitoring is consistent with our earlier results presented in Table 3 and is also both statistically and economically significant, since it corresponds to a 40% increase in profits after receiving VC financing.

In summary, these results empirically show the impact of monitoring on firm efficiency and productivity. We explicitly account for the endogenous nature of the VC firm selection process using a Heckman-style two-stage model, and after controlling for this selection, we find VCs have a significant positive monitoring effect on the post-financing TFP of the firms that they invest in.

### 4.2 Regression discontinuity analysis

In this section, we demonstrate the causal effect of venture capitalists on the productivity of the firms they invest in (i.e., monitoring) using a regression discontinuity (RD) approach. Specifically, we use the “fuzzy” RD technique discussed in Angrist and Lavy (1999) and Van der Klaauw (2002).[40] Identification in a fuzzy RD technique comes from a discontinuous jump in the probability of treatment. We use a two-stage approach similar to the ones used in the above articles to implement our RD analysis. Our estimation equation is

$$y_j = x_j\beta + \delta I_j + \varepsilon_j, \tag{5}$$

where $y_j$ is the growth in the productivity of the firm, $I_j$ is the endogenous treatment variable, which in our case is a dummy equal to one for VC-backed

[40] See also Frandsen (2008), Hahn, Todd, and Van der Klaauw (2001), and Lee and Lemieux (2010) for detailed discussions of the regression discontinuity design.

firms in the year they obtain VC financing and zero otherwise, and $x_j$ is a set of control variables. The fuzzy RD approach requires an assignment variable $S_j$ such that $\Pr(I_j|S_j)$ experiences a jump at a known point $S_j = S_0$. Note that, unlike the endogenous switching regression approach used earlier, $S_j$ can be correlated with the test variable $y_j$ (which is TFP growth in our case). The restriction required for identification is that no other effect causes a jump in $y_j$ at $S_j = S_0$. The regression discontinuity can be implemented with a two-stage approach. We first estimate

$$\Pr(I_j|S_j) = \phi.S_j + \gamma.1\{S_j \geq S_0\} + \eta_j, \tag{6}$$

where $1\{S_j \geq S_0\}$ is an indicator function that equals one if $S_j \geq S_0$ and zero otherwise, and $\eta_j$ is the error term. We augment the second-stage equation with $\Pr(I_j|S_j)$, which is calculated as the predicted probability from (6). The second-stage model is estimated as

$$y_j = x_j\beta + \delta \Pr(I_j|S_j) + \rho S_j + u_j. \tag{7}$$

This approach is analogous to a two-stage least-squares approach, with the discontinuity $1\{S_j \geq S_0\}$ serving as the identifying instrument.[41]

To obtain the discontinuity to identify our system, we exploit the fact that the U.S. Small Business Administration (SBA) provides financial support to firms up to a certain size cutoff.[42] For most industrial firms, the size cutoff is a prespecified level of employment that varies with the North American Industry Classification System (NAICS) industry affiliation of the firm.[43] Our assignment variable, $S_j$, is the normalized level of employment in the firm—i.e.,

$$S_j = \textit{Normalized Employment} = \frac{\textit{Firm Employment}}{\textit{Employment Cutoff for SBA Support Eligibility}}.$$

Thus, we expect a discontinuity in the propensity of firms to seek and obtain VC financing at $S_j = 1$. At all points below $S_j = 1$, firms are eligible for SBA support and therefore have a viable alternative to VC financing and are thus less likely to seek and obtain VC financing. At points above $S_j = 1$, firms are

41 The standard errors in the second stage are bootstrapped to account for generated regressors. We also conduct the above analysis in a two-stage least squares framework rather than estimating the first stage as a probit regression. In addition, we estimate a panel data two-stage least squares model with firm fixed effects. Our results do not qualitatively change in the above specifications.

42 The SBA provides a number of financial assistance programs for small businesses, including equity financing, debt financing through guarantees to commercial lenders to provide funding to small firms, and surety bond guarantees to help small businesses obtain contracts. In 2006, the total value of financial support provided by the SBA to small businesses was $78.1 billion, which grew to $90.45 billion in 2009, indicating that the SBA represents a significant source of financial support for smaller entrepreneurial firms. See http://www.sba.gov/idc/groups/public/documents/sba_homepage/serv_abtsba_2009_afr_001-030.pdf (page 7).

43 Size eligibility standards to receive SBA financial support are listed at the SBA website: http://www.sba.gov/idc/groups/public/documents/sba_homepage/serv_sstd_tablepdf.pdf.

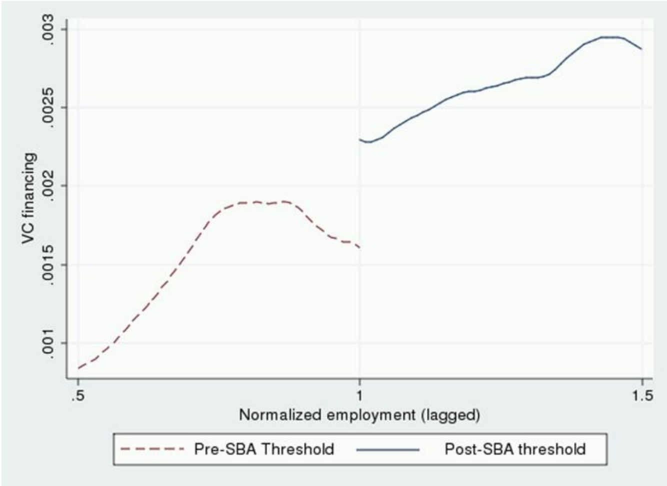


**Figure 1**
**Discontinuity of the treatment (VC financing) when normalized employment=1**

not eligible for SBA support, and as a result are more likely to seek and obtain VC financing.

We start by showing a graphical analysis of the discontinuity. Figure 1 graphs a kernel-weighted mean smoothing estimator of the propensity to obtain VC financing as a function of our assignment variable. The graph clearly indicates that there is indeed a jump in the propensity to obtain VC financing at the point where normalized employment equals one, as expected.

One concern with the assignment variable may be that firms may try to obtain federal funding by artificially restricting their size levels to comply with SBA requirements. McCrary (2008) argues that such active sorting may undermine the identification requirements of the regression discontinuity design. To rule out this concern, we graphically and statistically analyze the density function of the normalized employment variable around the cutoff point (i.e., at $S_j$=1). Visual analysis of the graph indicates a smooth density function of $S_j$ at the cutoff point. We then conduct the statistical test recommended in McCrary (2008) to ensure that our identification strategy is valid. The test is implemented as a Wald test of the null hypothesis that the discontinuity in the density function of the assignment variable is zero at the cutoff point. The test does not find a statistically significant jump or drop in the density function of the assignment variable $S_j$ at the cutoff point, i.e., $S_j = 1$, thereby validating our identification strategy.[44]

[44] The estimator for this test is calculated in two steps. First, one obtains a finely gridded histogram. Second, one smooths the histogram using local linear regression separately on either side of the cutoff point. The test statistic is based on the estimates of the density function from local linear regressions from the two sides of the cutoff point.

**Table 8**
**Regression discontinuity analysis for VC and non-VC-backed firms**

| First-stage | | Second-stage | |
|---|---|---|---|
| *Dependent Variable: VC Backing Dummy* | | *Dependent Variable: TFP Growth* | |
| Indicator (Norm. Emp. >1) | 0.357*** | VC Backing Prob. (predicted) | 1.325*** |
| | [0.046] | | [0.453] |
| Average 5 Year Prior TFP | 0.092** | Average 5 Year Prior TFP | −0.906*** |
| | [0.041] | | [0.009] |
| Norm. Employment | −0.001 | Norm. Employment | −0.001 |
| | [0.001] | | [0.001] |
| Firm Size | 0.153*** | Firm Size | −0.023*** |
| | [0.014] | | [0.003] |
| Number of Plants | 0.006*** | Number of Plants | 0.000 |
| | [0.001] | | [0.001] |
| Herfindahl Index | 0.307 | Herfindahl Index | −0.055** |
| | [0.19] | | [0.022] |
| Firm Age | −0.007*** | Firm Age | −0.005*** |
| | [0.002] | | [0.001] |
| Firm Market Share | 0.267 | Firm Market Share | 0.086 |
| | [0.341] | | [0.081] |
| Industry Risk | −5.616 | Industry Risk | 0.868* |
| | [4.543] | | [0.500] |
| High-Tech Firm | 0.595*** | High-Tech Firm | −0.023 |
| | [0.042] | | [0.02] |
| | | Firm Fixed Effects | Y |
| Year Fixed Effects | Y | Year Fixed Effects | Y |
| Observations | 404332 | Observations | 312885 |
| Wald $Chi^2$ | 1365.53*** | Adj. $R^2$ | 0.685 |

This table reports the result of a regression discontinuity analysis. The first stage reports the results from a probit model in which the dependent variable is whether a firm gets VC financing in a given year (*VC Backing Dummy*). The time series for each firm that gets VC financing terminates in the year of obtaining the first round of VC financing. The independent variables in the regressions are *Indicator (Norm. Emp. >1)*, which is a dummy variable that equals one if the one-year lagged ratio of the company's total employment to the maximum SBA size threshold to be eligible to receive support from the U.S. Small Business Administration (SBA) is greater than one, i.e., the firm is ineligible to obtain support from the SBA, and zero otherwise; *Norm. Employment*, which is the one-year lagged ratio of the company's total employment to the maximum SBA size threshold to be eligible to receive support from the U.S. SBA; *Average 5 year prior TFP*, which is the average TFP over the past five years starting from the current year; *Firm Size*, which is the one-year lagged value of natural log of the firm's capital; *Number of Plants*, which is the natural logarithm of the one-year lagged value of the number of plants in the firm; *Herfindahl Index*, which is the one-year lagged value of the measure of concentration of the firm's three-digit SIC industry; *Firm Age*, which is the natural logarithm one-year lagged value of the number of years since the firm first appeared in the LRD sample; *Firm Market Share*, which is the one-year lagged value of the firm's market share in terms of sales in the same three-digit SIC industry; *Industry risk*, which is the one-year lagged value of the median standard deviation of the total value of shipments calculated over a prior five-year period for all firms in the same three-digit SIC industry as the sample firm; *High-Tech Firm*, which is a dummy variable that takes the value one if the one-year lagged value of the firm's three-digit SIC code is 357, 366, 367, 372, 381, 382, or 384; and year dummies. The dependent variable in the second-stage regression is the *TFP Growth*, which is defined as the difference between the average TFP over the next five years and the average TFP over the past five years starting from the current year. The independent variables in these regressions are the *predicted VC backing probability* from the first stage and all the independent variables from the first stage except for *Indicator (Norm. Emp.>1)*. In the first-stage regression, heteroscedasticity corrected robust standard errors, which are clustered on firms, are in brackets, while in the second-stage regression bootstrapped standard errors, which are clustered on firms, are in brackets. All regressions are estimated with an intercept term. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

The results of the two-stage RD analysis are reported in Table 8. Consistent with our expectations, the coefficient estimate on the jump indicator (i.e., *Indicator(Norm.Emp.* $>$ 1)) is statistically significant and positive in the first-stage regression, indicating that there is a jump in the propensity to obtain VC financing at the point where a firm outgrows its eligibility for SBA support. In the second stage, we find that the coefficient estimate on the predicted probability of VC financing from the first stage is positive and statistically significant. Thus, after controlling for the selection of firms by VCs using the regression discontinuity approach, we find that VC financing leads to higher levels of productivity growth. Our results are therefore consistent with the notion that VC monitoring leads to an increase in firm productivity.

### 4.3 Propensity score matching

In this section, as an additional robustness check, we employ another methodology to confirm our earlier results on the effect of VC monitoring on subsequent growth in firm TFP. We employ a propensity score matching algorithm, where we match firms on multiple dimensions in the year prior to receiving VC financing. Specifically, we require that the matched firm be in the same three-digit SIC industry as the sample firm, be of similar size, and have a similar level of average prior TFP in the five years prior to receiving funding. The last criterion ensures that at the time of receiving VC financing, both our sample (VC-backed firms) and matched (non-VC-backed firms) portfolios have similar levels of productivity and efficiency.

Table 9 presents the results of the propensity score matching-based analysis. By construction, the average five-year TFP prior to receiving VC financing is similar and not statistically different between VC-backed and non-VC-backed matching firms. However, when we compare the TFP growth over the next five years for our two portfolios, we observe that while the VC-backed firms realize

**Table 9**
**Matched sample comparison of TFP growth after VC financing**

| | | Sample | Matched | Diff. |
|---|---|---|---|---|
| TFP Growth | Mean | 0.001*** | −0.047 | 0.048* |
| | *p*-value | (0.007) | (0.959) | (0.051) |
| | Quasi-median | 0.001*** | −0.040 | 0.041** |
| | *p* -value | (0.004) | (0.650) | (0.041) |
| | Observations | 558 | 558 | |
| Average 5 Year Prior TFP | Mean | 0.020 | 0.001 | 0.019 |
| | *p* -value | (0.949) | (0.341) | (0.460) |
| | Quasi-median | 0.040 | 0.010** | 0.030 |
| | *p* -value | (0.773) | (0.012) | (0.150) |
| | Observations | 588 | 588 | |

This table reports means and quasi-medians for *TFP Growth,* which is defined as the difference between the average TFP over the next five years and the average TFP over the last five years starting from the current year, and the *Average 5 Year Prior TFP,* which is the average TFP over the last five years starting from the current year. Quasi-medians are the average of the forty-third and the fifty-seventh percentiles for each variable. We report quasi-medians rather than medians to comply with the U.S. Census Bureau confidentiality requirements. The matched sample is created using a propensity score-based matching methodology. Matched firms are selected such that it is in the same three-digit SIC industry in the year of the VC financing of the sample firm and has comparable capital stock and average 5-year prior TFP as the sample firm. $P$-values are reported in parentheses. $P$-values for means and medians correspond to $t$-tests and sign-tests, respectively, for the null hypothesis that the mean and median are zero. $P$-values for differences in means and medians correspond to paired $t$-tests and sign-rank tests. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

significantly positive TFP growth (both at the mean and the quasi-median), the matched sample of non-VC-backed firms do not. Moreover, the difference in TFP growth between the two groups of firms is statistically and economically significant, with the mean of the VC-backed firms being approximately 5% higher and the quasi-median being approximately 4% higher than the matched sample of non-VC-backed firms. Consistent with our earlier findings, this result also suggests that VC involvement improves the efficiency of VC-backed firms through the extra-financial monitoring services provided by the VCs, with the magnitude of this effect being similar to that documented in the earlier sections.

One significant limitation of the propensity score matching analysis is that it does not control for the effect of unobservables on the selection of firms that get VC financing. Thus, we cannot entirely depend on the results in this section to demonstrate causality. We address the issue of unobservables using the switching regression and the regression discontinuity approaches discussed above.

## 5. The Channels Through Which VCs Improve Firm Efficiency

In this section, we identify the channels through which efficiency improvements are realized for VC-backed firms compared with non-VC-backed firms. In order to do this, we investigate the dynamics of firm output as well as the various inputs (capital, materials, and labor) around the years of receiving the first round of VC financing, benchmarked against that of non-VC-backed firms. We implement this using the regression specification outlined in Equation (3), where our dependent variables in the various regressions are as identified in the column headings in Table 10. We control for firm size and include both firm and year fixed effects and cluster the standard errors at the firm level. Panel A of Table 10 presents the regression results, while Panel B presents the changes in the dependent variables over time from before receiving VC financing to after receiving VC financing.

As can be seen from the results, sales of VC-backed firms are larger prior to VC financing and increase significantly over time from before receiving financing to after receiving financing as compared with that of non-VC firms. This increase in sales is even more pronounced in year five and after receiving the first round of financing. Similar to this increase in total sales, our results also document increases in total production cost for firms from before receiving VC financing to after receiving financing. This increase in total production cost mainly arises from increases in materials cost. Compared to non-VC-backed firms, VC-backed firms have higher total labor cost prior to receiving financing, and total labor cost for VC-backed firms also increases subsequent to receiving VC financing. While materials cost increases monotonically after receiving VC funding, the increase in labor cost is evident only during the first four years after receiving VC financing; for five years and

**Table 10**
**Dynamics of inputs to TFP around the first round of VC financing**

*Panel A: Regression Results*

| | Log Total Sales | Log Production Cost | Log Capital Exp. | Log Materials Cost | Log Salaries & Wages | Log Total Employment |
|---|---|---|---|---|---|---|
| VCBefore(–4,0) | 0.147*** | 0.113*** | 0.152*** | 0.104*** | 0.062** | 0.031 |
| | [0.032] | [0.036] | [0.051] | [0.037] | [0.028] | [0.029] |
| VCAfter(1,4) | 0.220*** | 0.180*** | 0.118** | 0.182*** | 0.112*** | 0.063* |
| | [0.043] | [0.044] | [0.059] | [0.043] | [0.034] | [0.034] |
| VCAfter(≥5) | 0.352*** | 0.293*** | 0.093 | 0.224*** | 0.103*** | 0.025 |
| | [0.051] | [0.049] | [0.060] | [0.049] | [0.039] | [0.042] |
| Firm Size | 0.593*** | 0.618*** | 1.040*** | 0.614*** | 0.581*** | 0.522*** |
| | [0.005] | [0.006] | [0.008] | [0.006] | [0.005] | [0.005] |
| Year Fixed Effects | Y | Y | Y | Y | Y | Y |
| Firm Fixed Effects | Y | Y | Y | Y | Y | Y |
| Observations | 546006 | 545941 | 552311 | 541722 | 546565 | 545239 |
| Adj. $R^2$ | 0.932 | 0.916 | 0.67 | 0.912 | 0.932 | 0.929 |

*Panel B: Change in Dependent Variable Over Time*

| | Log Total Sales | Log Production Costs | Log Capital Exp. | Log Materials Cost | Log Salaries & Wages | Log Total Employment |
|---|---|---|---|---|---|---|
| VCAfter(1,4) – VCBefore(4,0) | 0.072*** | 0.066** | −0.035 | 0.078*** | 0.050** | 0.032 |
| | [0.027] | [0.029] | [0.053] | [0.030] | [0.022] | [0.023] |
| VCAfter(≥5) – VCBefore(4,0) | 0.205** | 0.180*** | 0.059 | 0.120*** | −0.041 | 0.006 |
| | [0.037] | [0.036] | [0.057] | [0.038] | [0.031] | [0.033] |

This table reports results for panel data regressions where the dependent variables are log total sales, log production cost, log capital expenditure, log materials cost, log salaries and wages, and log total employment of a firm for a given year. The independent variables are *VCBefore(–4,0)*, which is a dummy variable that equals one for years –4 to 0 prior to obtaining the first round of VC financing and zero otherwise; *VCAfter(1,4)*, which is a dummy variable that equals one for years 1 to 4 after obtaining the first round of VC financing and zero otherwise; *VCAfter(≥5)*, which equals one in and after year 5 of VC financing and zero otherwise; *Firm Size*, which is the natural log of the firm's capital stock in a given year; and firm and year fixed effects. Panel A reports the regression coefficient estimates and their statistical significances. Panel B reports change in the dependent variable over various time periods and their statistical significances. Heteroscedasticity corrected robust standard errors, which are clustered on firms, are in brackets. All regression specifications are estimated with an intercept. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

after VC financing, we do not find a significant difference in the growth of salaries and wages from before receiving financing. We also find no changes in the level of total employment growth from before receiving VC financing to after receiving financing. Finally, we document that while capital expenditure is greater for VC-backed firms prior to receiving financing compared with non-VC-backed firms, there is no significant change in capital expenditure for VC-backed firms from before receiving financing to after receiving financing. Hellmann and Puri (2002) find that venture capitalists help professionalize firm management. Our results, particularly higher wages after VC backing, are consistent with their results, suggesting that VC backing leads to the employment of higher-quality workers in firms. Moreover, materials costs increase subsequent to VC backing.

Overall, the results presented in Table 10 suggest that the increase in efficiency of VC-backed firms that we document comes about through the improved product market performance of these firms (i.e., increases in sales), which may arise through the extra-financial services provided by the VCs. Further, the results are also consistent with the interpretation that VCs may be employing higher-quality workers in the years immediately after investing in the firm in order to improve their operating efficiency.

## 6. Differences Between High- and Low-reputation VC-backed Firms

### 6.1 Differences in TFP dynamics for high- and low-reputation VC-backed firms

Given the choice, entrepreneurs would prefer to source financing from high-reputation VCs rather than from low-reputation VCs. Hsu (2004) points out that a financing offer from a high-reputation VC is approximately three times more likely to be accepted by an entrepreneur and also that high-reputation VCs get better deal terms (i.e., lower valuations) when negotiating with startups. This suggests that startup firms will be willing to accept such terms only if high-reputation VCs provide superior monitoring and management, subsequently leading to better firm performance. In this section, our aim is to empirically show this. We thus run our fixed effects regression on the sample of high- and low-reputation VC-backed firms separately, and estimate the joint covariance matrix for (3) estimated for high- and low-reputation VC-backed firms.[45] In Panel A of Table 11, the first two columns correspond to high- and low-reputation VC-backed firm regressions, respectively; the third column presents the difference in TFPs between high- and low-reputation VCs

[45] We use a seemingly unrelated regression technique to compare the coefficients across the two equations estimated for high- and low-reputation VC-backed firms. In unreported regressions, we also estimated our results in a single regression, where we interacted the VC reputation dummy with our $VCBefore_{it}$ and $VCAfter_{it}$ variables. Our results remain qualitatively unchanged. We report these results because it is easier to interpret the coefficients and the differences between the coefficients in the two categories of VC reputation in this setup. Also, it provides for a more parsimonious display of the results.

at all stages; and the fourth column presents the difference in the net effect of monitoring (i.e., $TFP_{After} - TFP_{Before}$) between high- and low-reputation VCs.

Consistent with our earlier results, we find that both high- and low-reputation VCs actively engage in screening and select firms that on average have a higher TFP than non-VC-backed firms. Further, we find that low-reputation VCs screen firms with higher levels of initial TFP (averaged across five years prior to receiving financing) compared with high-reputation VCs. When we analyze the growth in firm TFP after receiving VC financing, we find that post-financing TFP growth is significantly greater for high-reputation VCs, suggesting that high-reputation VCs are better at improving productivity levels through monitoring compared with low-reputation VCs. We show that the above net effect of monitoring is both statistically and economically significant, on average being 10% higher for firms receiving funding from high-reputation VCs compared with firms receiving funding from low-reputation VCs. Our analysis presents an interesting and hitherto undocumented result, suggesting that most of the productivity gains of firms backed by low-reputation VCs can be attributed to the screening technology employed by such VCs, while in contrast, most of the productivity gains of firms that are backed by high-reputation VCs come from the ability of such VCs to improve the post-financing TFP of firms they invest in—i.e., through their monitoring activities. This is not surprising since high-reputation VCs have greater experience and expertise in managing entrepreneurial firms and therefore are able to provide additional extra-financial services to these firms that ultimately result in better operating efficiency and performance. The higher net effect of 10% on firm TFP arising due to better monitoring by high-reputation VCs translates to an increase in profits of approximately 35%, as mentioned previously. These results are also somewhat related with earlier empirical work by Gompers and Lerner (1999), who find that profit shares are higher for older and larger VCs. Similarly, these results are also related to those in Kaplan and Schoar (2005), who analyze both VC and buyout fund returns and show that there is a large degree of heterogeneity among fund returns. They show that returns tend to improve with the experience of the general partner. The above results establish that financing from reputable VCs (who are potentially also more experienced) lead to higher operating efficiency and TFP of their portfolio firms. Thus, when high-reputation VCs exit their investments it could potentially lead them to realize the higher returns that have been documented by the above articles.[46]

In Panel B of Table 11, we analyze the round-by-round improvements in firm TFP for firms backed by high- and low-reputation VCs. Consistent with

[46] We have also repeated these tests with another alternative measure of VC reputation, which is the market capitalization of IPO exits of the VC's portfolio firms over the prior five years as a fraction of IPO exit market capitalization of all VC-backed IPO exits over the prior five years (based on the measures used by Nahata 2008 and Ivanov, Krishnan, Masulis, and Singh 2010).Our results with this measure of VC reputation are qualitatively similar to those reported here.

**Table 11**
**TFP changes around year of VC financing for high- and low-reputation VC-backed firms**

*Panel A: TFP Regressions for High and Low Reputation Backed VCs*

| | High-reputation VC | Low-reputation VC | TFP Diff: (High–Low) | TFP Change Over Time Relative to VCBefore (–4,0): Diff.(High–Low) |
|---|---|---|---|---|
| VCBefore(–4,0) | 0.07*** | 0.081*** | −0.011*** | |
| | [0.017] | [0.016] | [0.001] | |
| VCAfter(1,4) | 0.174*** | 0.089*** | 0.085*** | 0.096*** |
| | [0.018] | [0.018] | [0.001] | [0.000] |
| VCAfter(≥5) | 0.262*** | 0.139*** | 0.122*** | 0.133*** |
| | [0.017] | [0.017] | [0.001] | [0.000] |
| Firm Size | −0.064*** | −0.064*** | | |
| | [0.001] | [0.001] | | |
| Herfindahl Index | −0.014 | −0.015 | | |
| | [0.011] | [0.012] | | |
| Year Fixed Effects | Y | Y | | |
| Firm Fixed Effects | Y | Y | | |
| Observations | 517718 | 518147 | | |
| Adj. $R^2$ | 0.413 | 0.413 | | |

*Panel B: TFP Regressions Over Rounds for High and Low Reputation Backed VCs*

| | High-reputation VC | Low-reputation VC | TFP Diff. | TFP Change Over Rounds Relative to VCBefore: Diff. (High–Low) |
|---|---|---|---|---|
| VCBefore | 0.055*** | 0.078*** | −0.023*** | |
| | [0.018] | [0.017] | [0.001] | |
| VCAfter(R1) | 0.120*** | 0.093*** | 0.027*** | 0.050*** |
| | [0.018] | [0.093] | [0.001] | [0.000] |
| VCAfter(R2) | 0.280*** | 0.127*** | 0.153*** | 0.176*** |
| | [0.026] | [0.029] | [0.001] | [0.001] |
| VCAfter(R3) | 0.282*** | 0.155*** | 0.128*** | 0.151*** |
| | [0.031] | [0.036] | [0.001] | [0.001] |
| VCAfter(≥R4) | 0.265*** | 0.073* | 0.192*** | 0.214*** |
| | [0.036] | [0.042] | [0.001] | [0.001] |
| Firm Size | −0.065*** | −0.064*** | | |
| | [0.001] | [0.001] | | |
| Herfindahl Index | −0.012 | −0.014 | | |
| | [0.012] | [0.012] | | |
| Firm Fixed Effects | Y | Y | | |
| Year Fixed Effects | Y | Y | | |
| Observations | 517130 | 517707 | | |
| Adj. $R^2$ | 0.413 | 0.412 | | |

This table reports results for panel data regressions in which the dependent variable is the TFP of a firm for a given year. The regression results are reported separately for high- and low-reputation VC-backed firms. The independent variables in Panel A are *VCBefore(–4,0)*, which is a dummy variable that equals one for years –4 to 0 prior to obtaining the first round of VC financing and zero otherwise; *VCAfter(1,4)*, which is a dummy variable that equals one for years 1 to 4 after obtaining the first round of VC financing and zero otherwise; *VCAfter(≥5)*, which equals one in or after year 5 of VC financing and zero otherwise; *Firm Size*, which is the natural log of the firm's capital stock in a given year; *Herfindahl Index*, which is the concentration of the firm's three-digit SIC industry; and firm and year fixed effects. The independent variables in Panel B (not included in Panel A) are *VCBefore*, which is a dummy variable that equals one for all years prior to the first round of VC financing and zero otherwise; *VCAfter(R1)*, which is a dummy variable that equals one for years after the first round but before the second round of VC financing and zero otherwise; *VCAfter(R2)*, which is a dummy variable that equals one for years after the second round but before the third round of VC financing and zero otherwise; *VCAfter(R3)*, which is a dummy variable that equals one for years after the third round but before the fourth round of VC financing and zero otherwise; *VCAfter(≥R4)*, which is a dummy variable that equals one for years after the fourth round of VC financing and zero otherwise. Heteroscedasticity corrected robust standard errors, which are clustered on firms, are in brackets. All regressions are estimated with an intercept term. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

the results above, we first show that the TFP of firms backed by low-reputation VCs is significantly greater than the TFP of firms backed by high-reputation VCs prior to receiving VC financing, suggesting that low-reputation VCs screen firms with higher initial levels of TFP. However, the results also show that after each round of receiving VC financing, the TFP of firms backed by high-reputation VCs is significantly greater than the TFP of firms backed by low-reputation VCs, again suggesting that high-reputation VCs are better able to increase the post-financing efficiency of firms due to their better monitoring abilities. The results on the round-by-round improvement in firm TFP for high- and low-reputation VCs show that the TFP improvement after VC backing is much greater for firms backed by high-reputation VCs than for those backed by low-reputation VCs.

Overall, our results are consistent with higher-reputation VCs providing greater monitoring services to the firms that they back, since these firms experience greater improvements in their productivity subsequent to VC financing.[47]

### 6.2 Differences in channels of TFP improvements between high- and low-reputation VC-backed firms

In this section, we investigate if there are differences in the underlying process of efficiency improvements between firms backed by high-reputation VCs and those backed by low-reputation VCs. We also present evidence on how greater efficiency (TFP) improvements are realized by high-reputation VCs. We do so by estimating the joint covariance matrix for (3) estimated for high- and low-reputation VC-backed firms as in Table 11. In Table 12, we investigate the dynamics of total sales and total production costs around the year of receiving the first round of VC financing for firms backed by high- and low-reputation VCs, benchmarked against that of non-VC-backed firms.

Panel A of Table 12 presents the results for total sales for high- and low-reputation VC-backed firms benchmarked against non-VC-backed firms. The results show that prior to receiving financing, total sales for both high- and low-reputation VC-backed firms are significantly greater than that of non-VC-backed firms. Moreover, at this time, total sales for firms backed by high-reputation VCs are greater than that of firms backed by low-reputation VCs, with the difference being on average around 2%. Subsequent to receiving VC financing, we also find that the growth in sales (from before to after receiving financing) is significantly greater for firms that are backed by high-reputation VCs compared with those backed by low-reputation VCs with

47 In unreported tests, we also run the switching regression and the counterfactual analyses from Table 7 to control for the endogenous matching between entrepreneurial firms and high- and low-reputation VCs. We find that TFP growth would have been smaller for high-reputation VC-backed firms had they received financing from low-reputation VCs instead. Similarly, TFP growth would have been greater for low-reputation VC-backed firms had they received financing from high-reputation VCs. In addition, we also conduct a propensity score matched analysis comparing propensity score matched sample adjusted TFP growth of high- and low-reputation VC-backed firms. We find significantly greater post-financing matched sample adjusted TFP growth for high-reputation VC-backed firms relative to low-reputation VC-backed firms.

**Table 12**
**Dynamics of inputs in TFP around the first round of VC financing for high- and low-reputation VC-backed firms**

*Panel A: Log Total Sales*

| | High-reputation VC | Low-reputation VC | Diff. (High–Low) | Diff. in growth relative to VCBefore(–4,0): (High–Low) |
|---|---|---|---|---|
| VCBefore(–4,0) | 0.161*** | 0.141*** | 0.019*** | |
| | [0.091] | [0.018] | [0.001] | |
| VCAfter(1,4) | 0.250*** | 0.205*** | 0.045*** | 0.025*** |
| | [0.020] | [0.020] | [0.001] | [0.001] |
| VCAfter(≥5) | 0.379*** | 0.362*** | 0.017*** | −0.002 |
| | [0.019] | [0.019] | [0.001] | [0.002] |
| Firm Size | 0.568*** | 0.567*** | | |
| | [0.001] | [0.001] | | |
| Year Fixed Effects | Y | Y | | |
| Firm Fixed Effects | Y | Y | | |
| Observations | 538216 | 538627 | | |
| Adj. $R^2$ | 0.932 | 0.932 | | |

*Panel B: Log Production Cost*

| | High-reputation VC | Low-reputation VC | Diff. (High–Low) | Diff. in growth relative to VCBefore(–4,0): (High–Low) |
|---|---|---|---|---|
| VCBefore(–4,0) | 0.109*** | 0.112*** | −0.003** | |
| | [0.024] | [0.022] | [0.001] | |
| VCAfter(1,4) | 0.171*** | 0.189*** | −0.018*** | −0.015*** |
| | [0.025] | [0.024] | [0.001] | [0.001] |
| VCAfter(≥ 5) | 0.267*** | 0.345*** | −0.078*** | −0.075*** |
| | [0.023] | [0.024] | [0.001] | [0.001] |
| Firm Size | 0.586*** | 0.586*** | | |
| | [0.005] | [0.001] | | |
| Year Fixed Effects | Y | Y | | |
| Firm Fixed Effects | Y | Y | | |
| Observations | 538150 | 538562 | | |
| Adj. $R^2$ | 0.917 | 0.917 | | |

This table reports results for panel data regressions where the dependent variables are log total sales and log production cost for a given year. The regression results are reported separately for high- and low-reputation VC-backed firms. The independent variables are *VCBefore(–4,0)*, which is a dummy variable that equals one for years –4 to 0 prior to obtaining the first round of VC financing and zero otherwise; *VCAfter(1,4)*, which is a dummy variable that equals one for years 1 to 4 after obtaining the first round of VC financing and zero otherwise; *VCAfter(≥ 5)*, which equals one in and after year 5 of VC financing and zero otherwise; *Firm Size*, which is the natural log of the firm's capital stock in a given year; and firm and year fixed effects. Heteroscedasticity corrected robust standard errors, which are clustered on firms, are in brackets. All regressions are estimated with an intercept term. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

the difference in sales growth being on average around 2.5%. Thus, the larger improvement in TFP achieved by high-reputation VC-backed firms, documented in Table 11, could partially be explained by this better product market performance through increased sales growth of high-reputation VC-backed firms compared with low-reputation VC-backed firms.

Panel B of Table 12 presents the results for the total production costs of high- and low-reputation VC-backed firms. We find that total production cost is consistently lower for high-reputation VC-backed firms compared with low-reputation VC-backed firms both before and after receiving VC financing, even though it is significantly above that of non-VC-backed firms. Moreover, the increase in production cost from before receiving financing to after receiving financing is also lower for firms that are backed by high-reputation VCs, suggesting that active monitoring by high-reputation VCs may lead to lower increases in cost for these firms, thus leading to more efficient production and higher productivity gains for high-reputation VC-backed firms compared with low-reputation VC-backed firms. This lower increase in total production cost for high-reputation VC-backed firms is on average around 1.5% in the first four years after receiving financing and around 7.5% in years five and after, both of which are statistically and economically significant.[48]

The results presented in this section clearly show us how high-reputation VC-backed firms are able to achieve higher increases in productivity and efficiency improvements compared with firms backed by low-reputation VCs. It also shows us how better monitoring abilities of high-reputation VCs may be reflected through the production technology. We establish that the higher improvements in TFP and efficiency achieved by high-reputation VC-backed firms come from both better product market performance by such firms, through higher sales realizations, as well as through various cost reductions associated with the production process. These results therefore attest to the better monitoring ability of high-reputation VCs, which are able to achieve better sales using lower input levels and thus are able to attain higher levels of productivity improvements for the firms they invest in.

## 7. Impact of VC Backing and Productivity on the Probability of Exit

In this section, we analyze the effect of VC financing on the exit probability of firms and whether the increased productivity of VC-backed firms leads to a higher probability of a successful exit. We relate VC backing and the TFP of a firm to its probability of exit either through an IPO or through a merger/acquisition (M&A) as opposed to a write-off. We analyze the effect of VC backing and TFP growth on the probability of a successful exit using a multinomial logit model. We estimate the following multinomial logit model:

$$Exit\_Type = F(VC\ backing, TFP\_Growth, Avg\_prior\_TFP, Firm\_size, Controls), \tag{8}$$

[48] In unreported tests, we investigate other components of input—i.e., materials, labor, and capital. We find a similar pattern of a lower change in materials cost and capital expendiure (from before to after) for high-reputation VC-backed firms compared with low-reputation VC-backed firms. We also find that the labor cost increases more for high-reputation VC-backed firms in the short run and less for high-reputation VC-backed firms in the long run, compared to low reputation VC-backed firms. This is consistent with high-reputation VCs initially employing more skilled labor, as suggested by Hellmann and Puri (2002).

**Table 13**
**Venture capital backing and IPO and acquisition frequency**

| | IPO | M&A |
|---|---|---|
| VC Backing | 1.119*** | 1.655*** |
| | [0.184] | [0.153] |
| TFP Growth | 0.360*** | 0.024 |
| | [0.120] | [0.094] |
| Average Prior TFP | 0.292** | 0.037 |
| | [0.130] | [0.097] |
| Firm Size | 0.434*** | 0.305*** |
| | [0.028] | [0.027] |
| Firm Age | −0.031*** | 0.003 |
| | [0.006] | [0.005] |
| Herfindahl Index | 0.231 | 0.525 |
| | [0.510] | [0.540] |
| Observations | 405798 | |
| *Chi*$^2$ | 216807.7 | |

This table reports the results of multinomial logit estimation with type of exit (i.e., No Exit, IPO, or M&A) as the dependent variable. *No Exit* is the base case outcome. The dependent variables are *VC backing*, which is a dummy variable that is one for firms that obtain VC financing; *TFP Growth*, which is the TFP Growth calculated as current year TFP minus the prior five-year average TFP; *Average Prior TFP*, which is the five-year average TFP prior to the current year; *Firm Size*, which is the natural log of the firm's capital stock in the current year; *Herfindahl Index*, which is the measure of concentration of the firm's three-digit SIC industry; *Firm Age*, which is the natural logarithm of the number of years since the firm first appeared in the LRD sample; and year and two-digit SIC industry dummies. Heteroscedasticity corrected robust standard errors, which are clustered on firms, are in brackets. The regression is estimated with an intercept term. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

where *Exit_Type* is a dummy variable representing three categories: write-offs (the base category), M&As, and IPOs; *Avg_prior_TFP* signifies the five-year average TFP prior to the current year; and TFP_Growth is the current year TFP minus the average prior five-year average TFP. In addition to firm size, we also control for firm age, Herfindahl index of the firm's industry, year dummies, and two-digit SIC industry dummies. Our results estimating (8) are presented in Table 13.

We find that VC backing has a significant positive effect on the exit probability (for both IPO and M&A exits). Further, both TFP growth and prior average TFP have a positive relation with the exit probability through IPOs, but not with the exit probability through acquisitions. In addition, we also find that firm size is positively related to the probability of a successful exit through either an IPO or an M&A, while younger firms are more likely to exit through an IPO.

Overall, the results suggest that VC backing has a positive impact on IPO and M&A exits, and further, TFP growth has a positive and significant association with the probability of IPO exits.

## 8. Relation to the Existing Literature

Our article is related to the existing literature that compares VC-backed firms with non-VC-backed firms. Some important articles in this literature are Hellmann and Puri (2000, 2002) and Hsu (2006). Hellmann and Puri (2000) provide evidence that VC financing is related to the product market strategies and outcomes of startups, while Hellmann and Puri (2002) provide evidence regarding the professionalization of startup firms. Hsu (2006) analyzes the impact of VC backing on the frequency of cooperative commercialization strategies of technology-based startups. He finds substantial boosts in cooperative activity associated with VC-backed firms. We extend the above literature by comparing the overall efficiency of VC-backed and non-VC-backed private firms, and by analyzing *how* and *when* efficiency improvements arise due to VC backing.

Other studies in the literature have focused on the monitoring role of VCs based on samples of VC-backed firms; see, e.g., Lerner (1995), who examines VCs' representation on the boards of private firms and analyzes whether this representation is greater when the need for oversight is greater. We extend this literature by distinguishing between the screening and monitoring roles of venture capitalists, directly comparing samples of VC-backed and non-VC-backed firms, as well as high-reputation VC-backed and low-reputation VC-backed firms.[49] In addition, Bottazzi, Da Rin, and Hellmann (2008) find that VC activism is related to VC partner experience and VC firm characteristics (such as whether the VC firm is independent or related to another institution such as a company, bank, or government-owned agency). Finally, the effect of VC backing on IPOs has been documented by several articles. Megginson and Weiss (1991) study the certification role of venture backing in the financing of IPOs. More recently, using a sample of venture-backed firms, Sorensen (2007) shows that companies funded by more experienced VCs are more likely to go public. He documents that this follows both from the direct influence of more experienced VCs and also from sorting in the market for venture capital. While our results are broadly consistent with the evidence presented in his article, we analyze the productivity of VC-backed firms, while Sorensen (2007) analysis uses only IPO exits as a measure of performance. Our article also contributes to the emerging literature on the role of financial intermediaries in the birth, growth, and death of entrepreneurial firms: see, e.g., Kerr and Nanda (2009).[50]

In a contemporaneous related article, Puri and Zarutskie (2009) study the life-cycle dynamics of VC-backed and non-VC-backed firms and examine the importance of VCs in new firm creation. Their results show that VCs typically invest in firms that exhibit potential for large scale prior to exit. They also analyze the survival of VC-backed firms relative to that of non-VC-backed

49 Our article is also related to Gompers and Lerner (1999), who find that profit shares are higher for older and larger VCs, and Kaplan and Schoar (2005), who analyze both VC and buyout fund returns and show that there is a large degree of heterogeneity among fund returns and returns tend to improve with the experience of the general partner.

50 Our article is also related to the literature on firms' choice between private and public sources of financing; see, e.g., Spiegel and Tookes (2008) and Chemmanur and Fulghieri (1999). To the extent that we also study the propensity of venture-backed firms to exit through IPOs, our article is also related to the broader literature on IPOs. See Ritter and Welch (2002) for a review of this literature.

firms. In contrast, we focus on distinguishing between the screening and monitoring effects of VC backing, how such activities affect firm productivity, and the differences between high- and low-reputation VCs in accomplishing these activities. Further, we document several new results related to the channels through which VCs improve entrepreneurial firm productivity, and the relationship between such productivity improvements and the probability of successful exit.

## 9. Conclusion

We use the Longitudinal Research Database (LRD) of the U.S. Census Bureau, which covers the entire universe of private and public U.S. manufacturing firms, to study several related questions regarding the efficiency gains generated by venture capital (VC) investment in private firms. Our analysis shows that the overall efficiency of VC-backed firms is higher than that of non-VC-backed firms at every point in time. This efficiency advantage of VC-backed firms arises from both screening and monitoring: The efficiency of VC-backed firms prior to receiving financing is higher than that of non-VC-backed firms, and further, the growth in efficiency subsequent to VC financing is greater for such firms. The above increases in efficiency of VC-backed firms are spread over the first two rounds of VC financing, after which the TFP of such firms remains constant until exit. We further show that while the TFP of firms prior to receiving financing is lower for high-reputation VC-backed firms than for low-reputation VC-backed firms, the increase in TFP subsequent to financing is significantly greater for these firms, consistent with high-reputation VCs having greater monitoring ability.

We disentangle the screening and monitoring effects of VC backing using three different methodologies: switching regression with endogenous switching; regression discontinuity analysis; and propensity score matching. We show that while overall efficiency gains generated by VC backing arise primarily from improvements in sales, the efficiency gains of high-reputation VC-backed firms arise also from lower increases in production costs. Finally, we show that VC backing and the associated efficiency gains positively affect the probability of a successful exit.

## Appendix A: Switching Regression Model

A switching regression model with endogenous switching consists of a binary outcome equation that reflects the selection or matching between the VC and the firm, and two regression equations on the variable of interest, in this case *TFP growth*. Formally, we have

$$I_i^* = Z_i'\gamma + \varepsilon_i, \tag{9}$$

$$y_{1i} = x_i'\beta_1 + u_{1i}, \text{ and} \tag{10}$$

$$y_{2i} = x_i'\beta_2 + u_{2i}. \tag{11}$$

Equation (9) is the latent VC-firm matching equation. To reflect binary outcomes, $I^*$ is discretized as follows:

$$I_i = 1 \text{ iff } I_i^* > 0, \text{ and } I_i = 0 \text{ iff } I_i^* \leq 0. \tag{12}$$

In other words, $I_i$ equals one if and only if a firm receives VC financing. In this setup, the VC firm matching is modeled in reduced form. The dependent variable $I_i$ indicates the outcome of whether a firm receives VC financing, which results from decisions of both the firm and the VC and the screening technology adopted by the VC. Accordingly, in the empirical specification, the vector $Z_i$ contains variables that are related to the VC or the entrepreneurial firm. We estimate this first-stage equation using a dynamic probit model in which the dependent variable is a binary dummy, identifying whether a firm receives VC backing or not.[51]

Equation (10) analyzes the growth in TFP after receiving financing for VC-backed firms while controlling for selection effects, while Equation (11) does the same for firms that did not receive VC financing. To estimate the model, a key observation is that since either Equation (10) or (11) is realized depending on the outcome of $I^*$ (but never both), the observed TFP growth is a conditional variable. Taking expectations of Equation (10), we obtain

$$\begin{aligned} E[y_{1i}] &= E[y_i | I_i = 1] \\ &= E[y_i | I_i^* > 0] \\ &= E[X_i^{\prime}\beta_1 + u_{1i} | Z_i^{\prime}\gamma + \varepsilon_i > 0] \\ &= X_i^{\prime}\beta_1 + E[u_{1i} | \varepsilon_i > -Z_i^{\prime}\gamma]. \end{aligned} \tag{13}$$

Because $u_1$ and $\varepsilon$ are correlated, the last conditional expectation term in (13) does not have a zero mean, and OLS on Equation (10) will generate inconsistent estimates. If, however, Equation (10) is augmented with the inverse Mills ratio from the first-stage probit estimation as a right-hand-side variable, we can use OLS to find consistent estimates. This procedure is discussed in detail in Heckman (1979) and Maddala (1983). Equation (9) is first estimated by a probit regression, yielding consistent estimates of $\gamma$. With this, the inverse Mills ratio terms can be computed for Equations (10) and (11). Both equations are then augmented with the inverse Mills ratios as additional regressors.[52] These terms adjust for the conditional mean of $u$, and allow the equations to be consistently estimated by OLS.

One can compute the hypothetical TFP growth for VC-backed firms had they not received VC financing using Equation (11) and the hypothetical TFP growth for non-VC-backed firms had they received VC financing using Equation (10). Thus, to infer the monitoring effects of VCs on firm TFP growth, we compute the following difference:

$$\underbrace{y_{1i}}_{\text{actual}} - \underbrace{E[y_{2i} | I_i^* > 0]}_{\text{hypothetical}}. \tag{14}$$

The first term in (14) is the actual TFP growth of a VC-backed firm, while the second is the hypothetical TFP growth that would be obtained by the same firm had it not received VC financing. If the difference is positive, then the impact on TFP growth due to the monitoring services provided by the VC is explicitly quantified, since the actual TFP growth achieved by VC-backed firms is higher.

51 The dependent variable is always equal to zero for all non-VC-backed firms. For VC-backed firms, it is zero in all years prior to receiving VC financing, and it equals one in the year the firm receives VC financing. It is set to missing in the following years. Thus, in the first stage, VC-backed firms effectively drop out of the sample for all years subsequent to the year of receiving financing.

52 The standard errors in the second stage are bootstrapped to account for generated regressors.

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